
What Moves the Ceiling on Automated Audit: from General Code to Scientific Code, Results and Data

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Abstract

Models increasingly check what other models write. We measure one shipped arrangement against ground truth no model supplies: an auditor from a different vendor reads the work, and deterministic checks verify what code can verify. On general-purpose code with hidden tests, at least one of eight independent readings flagged 30.0% [20.0, 40.7] of defective programs that pass their visible tests, and 16.0% [10.1, 22.3] of correct ones, with diminishing returns and no plateau. That rate belongs to a configuration, not to the auditor: the auditor route (model and sampling together), one rulebook sentence and the counting rule each move it, and on a post hoc rating most missed programs fail where the specification's prose does not entail the value the hidden suite expects. Three preregistered studies then examine scientific code, reported results and data. Checks that every reported number traces to its source flagged every inconsistency and no fabrication written consistently into log, results and report; the model auditor flagged nearly all such fabrications, twice on a wrong calculation (on post hoc inspection). On data, a documentation-derived validator and the model flagged overlapping but different faults, so together they covered more than either; on held-out data, at a clean-flag budget fixed in advance and with diagnosis scored by code, adding the model to the validator raised correctly diagnosed faults by 10.7 points [6.0, 15.9]. On scientific code with no tests shown, reframing the task around one step sharply reduced flags on correct programs. Where both tiers ran, every flag on a clean item and every wrong finding came from the model, partly by construction of the clean items. Studies making model calls were preregistered, the two unregistered analyses are named, and every quoted report passed indepen-

dent cross-vendor review.

1. Introduction

Here is a program that none of twenty readings, by three different auditor models, flagged. The task, MBPP+ problem 102 (Liu et al., 2023), asks for a function that converts a snake-case string to camel case, and shows three ordinary examples. The generated function splits on underscores and capitalises each word, and it passes all three. The hidden suite then passes `__python_program` and expects `__PythonProgram`; the function returns `PythonProgram`. The task does not say how leading underscores should be handled. Both model raters in a post hoc assessment classified this instance as undetermined by the prose, and it still counts as defective under our hidden-test definition.

When a model writes code, reports a result or prepares a dataset, someone has to check it, and increasingly that someone is another model. This paper measures how often such a checker flags defective work, what moves that rate, and whether it helps where checking matters most, in science. We study one shipped system, CrossAudit (Section 2), through which every auditor reading here passes. It rests on two rules: the model that audits comes from a different vendor from the model that wrote the work, and whatever a deterministic check can verify is verified by code, not by a model's opinion. The shipped configuration pairs different vendors; our comparisons also include same-vendor controls.

An auditor's score on a benchmark combines at least four things: whether it flags a program, which of its findings count as blocking, how many times it is asked, and whether the expected behaviour follows from the specification it is shown. Work on critic models usually reports the aggregate. We vary these factors one at a time where the design allows, against a truth no model supplies: a hidden test suite run by an interpreter. Two limits apply throughout. Changing the reading model also changed its sampling, so route comparisons do not isolate model ability. And the role of the specification is a post hoc association, not a tested cause.

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General-purpose code (Section 3). At least one of eight independent readings of the shipped auditor flags 30.0% [20.0, 40.7] of defective programs that pass their visible tests, and 16.0% [10.1, 22.3] of correct ones. Each extra reading adds less, but the eighth still adds 1.93 points [1.14, 2.78]: no plateau within eight readings. By the *ceiling* of a configuration we therefore mean the union recall it reaches within the readings we could afford, at the false-positive rate that cost, not an asymptote, which would be an extrapolation. The rate also changes with the auditor route, its instructions and the counting rule. What no reading flags mostly resembles the example above: of the 57 defective programs none of those twenty readings flagged, two model raters judge that in 44 (77.2% [62.1, 91.1], post hoc) the prose does not entail the value the hidden suite expects on the first failing input: it is silent, open to the program’s reading, or contradicts the suite.

Science (Section 4). Three preregistered studies examine the jobs a scientist would hand an auditor: scientific code, reported results and data. The results and data studies compare model auditing with deterministic verification, and the two failed in different places. Checking that every reported number traces to its source flagged every inconsistency among a report, its results file and its run log, and none of the results that were wrong consistently in all three; the model auditor flagged nearly all of those. On data, a validator written from the documentation and the model flagged overlapping sets of faulty items, so together they covered more than either. On held-out items, with operating points fixed in advance at a clean-flag rate of at most 10% and diagnosis decided by code, adding the shipped auditor to the validator raised correctly named faults by 10.7 points [6.0, 15.9] (preregistered). On scientific code, with no tests shown, a task reframing that specified the step as the deliverable, moved the overall problem into context and exempted earlier functions sharply reduced the auditor’s flags on correct programs, and less sharply on defective ones.

Contributions.

- A way to measure an automated auditor’s limit against executed ground truth, with union recall always reported beside the false-positive rate it cost: controlled changes to the number of readings and the rulebook, route comparisons that change model and sampling together, re-grading under decision rules, and a post hoc association with what the specification determines. On general-purpose code, each controlled or route contrast moves the flag rate, so a recall figure without its operating point does not transfer; the specification analysis describes an association, not a tested cause.
- Operating points of the model auditor on scientific code, and comparisons of the model tier with deterministic verification on reported results (the shipped

provenance checks and re-execution) and on data (a documentation-derived validator). The tiers flag different faults, and each fails where the other does not.

- A lesson for evaluating auditors. In the results and data studies, reading every flag on a clean item showed the auditor finding real defects in our own construction of those items. An evaluation must establish that its clean items are clean by the auditor’s standard; otherwise its apparent false-positive rate can include valid findings about defects in the evaluation materials.

Preregistration and review. Every study that made model calls was preregistered before its first call, with hypotheses, intervals and stopping rules fixed in advance and departures recorded as numbered amendments. Two whole analyses were not registered, and the paper names both (Appendix A); every other unregistered quantity is labelled post hoc where it appears. Five preregistered hypotheses met their own rejection criteria, one of them in the science studies. Every report was reviewed against the raw records by a model from a different vendor, and the reviews are archived verbatim (Appendix A). Section 5 discusses what the measurements mean. The full method, results and discussion are in Appendices A–D, and the records, reviews and the command that regenerates every figure are described in Appendix G.

2. The system under study

Every auditor reading in this paper is taken through one shipped system, CrossAudit. Every auditor prompt is built by the product’s own prompt builder, and every family but one is read through the product’s own audit call; the exception, `gpt-6-astra`, receives the same prompt bytes through a direct call. We describe the system first because its design fixes what the later sections can and cannot vary. We state its structure only here and quote no measurement of it. Its limits are the subject of Section 3 onward.

Separation of roles. A generator model writes an increment (a set of files) into a Git repository. An auditor model then reads the committed bytes and returns findings, each graded BLOCKER or ADVISORY. The auditor must come from a different vendor from the generator: the audit refuses to run when the configuration names the same vendor for both roles, and when the generator’s vendor is not declared, the receipt records that independence could not be asserted. The same-vendor arms below run with the generator’s vendor undeclared, deliberately. The premise is that a model reviewing its own vendor’s output is an interested reader. The product takes that premise as given, and Section 3 puts one consequence of it to the test: whether a same-vendor auditor catches more or less.

Two tiers of evidence. Before the auditor reads anything, a deterministic check layer runs registered checks over the

same committed bytes. A check’s finding is *verified*: code produced it and code can reproduce it. A model’s finding is *raised*: it is a reading, not yet reproduced. The checks are grouped into named profiles, so strictness is a setting the project chooses:

- **general**, the default: structured files parse, every local path a file lists as an input or dependency exists, and relative links and leftover placeholders are reported as advisory.
- **science**: results carry a schema; quantities carry units; a result that reports convergence must meet its own threshold; every input a result depends on is listed, exists and is cited at an exact revision (*provenance*); and every number a report states is annotated with the file and quoted span it came from, which a check matches against the cited bytes (*number*→*source*).
- **research**: the general profile’s checks plus a citation check under which a report may cite only sources it fetched through a governed tool.

The science profile follows one rule throughout, and the rule is the one Section 4 tests: *the model names its evidence, and code verifies it*. A number with no source line, a unit that disagrees with its schema or an input that does not exist fails without any model’s opinion.

Computed verdicts. The verdict is derived by code from the evidence, in a fixed order. A failed deterministic check blocks, whatever the model says. A model BLOCKER with no deterministic support either blocks or escalates to a person, according to a project setting. A blocked increment goes back to the generator for a bounded number of revision rounds, three by default. Each revision passes a repair guard before it is committed: an edit outside the audited scope is refused, and a new catch-all handler, a deleted assertion or a skipped test is flagged to the auditor for the next round.

Receipts. A passing round produces a receipt that binds the verdict to the commit that was audited, the commit of the rulebook the auditor was given, and a digest over every finding with its tier. Every model call is written to a ledger with its prompt hash and cost. A verification command recomputes the digest from the recorded evidence, so a receipt can be checked by someone who does not trust the operator.

What the rest of the paper measures. Section 3, with Appendices A and B, measures the model tier on its own. There, an increment counts as flagged when the auditor grades any finding BLOCKER, and the general profile’s checks run as in the product but decides no recall or false-positive outcome. The model tier is the part of the system whose limit is least known, since a deterministic check catches exactly what it was written to catch. Section 4

measures the model tier on scientific code, and sets it beside deterministic verification on reported results and on data, asking what each flags that the other does not.

3. The ceiling on general-purpose code

Substrate and ground truth. HumanEval+ and MBPP+ (Liu et al., 2023): Python functions with an executable hidden test suite. One frozen generator (claude-haiku-4-5) was shown each specification and its visible tests and wrote one solution per problem, over two passes of 540 problems. A candidate that passes the visible suite and fails the hidden one is *defective*; one that passes both is *correct*. Recall is measured on all 110 defective instances (56 problems) and false positives on 150 correct ones drawn by seed. The auditor sees the specification, the visible tests and the candidate, never the hidden suite. Because the defective stratum already survived the visible tests, recall here is the fraction of that harder subset caught, not of all defects a model writes (Appendix A).

Estimand and statistics. Union recall at K : the fraction of defective instances blocked (any finding graded BLOCKER) by at least one of K independent readings, averaged exactly over all size- K subsets of the readings. The same union on correct instances is the false-positive rate, always reported beside it (Table 1 collects the primary quantities). Intervals are 95% problem-cluster percentile bootstraps; paired contrasts carry exact McNemar and cluster sign-flip tests. The shipped cross-vendor route (gpt-5.6-terra) sends no sampling parameter; same-vendor routes are sent temperature 0, a difference that bounds every cross-family union contrast (Section B.2).

Repeated reading. Union recall rises from 10.7% at one reading to 30.0% [20.0, 40.7] at eight, at 16.0% [10.1, 22.3] on correct code (Figure 1). The eighth reading still adds 1.93 points [1.14, 2.78], above the 1.0-point flattening bar registered in advance, so we report diminishing returns, not saturation. Twenty readings pooled across three families reach 48.2% [36.7, 60.0] at 36.0% [28.2, 44.2] on correct code: pooling buys coverage at its own price.

Which model reads. Replacing the auditor with a stronger same-vendor model lowered union recall at eight readings by 26.4 points [−37.3, −15.6] (exact McNemar $p = 4.2 \times 10^{-7}$; cluster sign-flip $p = 2.0 \times 10^{-5}$), a preregistered primary. This is not a measure of auditing ability. The stronger model also flags far less correct code, 3.3% [0.7, 6.7] against 16.0%, and under an exploratory any-finding rule the sign reverses: 59.1% [47.3, 70.6] recall at 34.0% [26.5, 41.7] on correct code, against 34.5% [23.6, 45.5] at 19.3% [12.8, 26.3] (Figure 3 in Appendix B). At one reading, before the differently sampled unions grow apart, the gap is only −10.1 points. A high-reasoning cross-vendor model

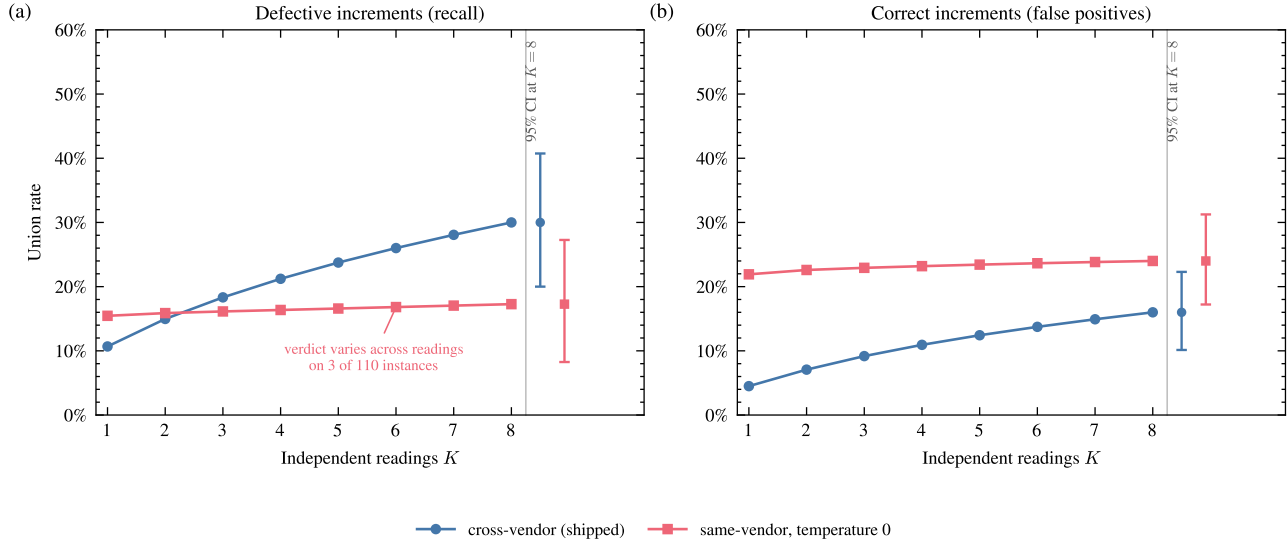


Figure 1. Returns to repeated reading diminish short of complete. Union rate against the number of independent readings K , for two auditor families. (a) Recall on defective instances; (b) false positives on correct instances. Bars at the right of each panel give the 95% problem-cluster bootstrap interval of each series at $K = 8$, dodged in x so overlapping intervals stay separable. The cross-vendor curve does not meet the preregistered flattening bar (its last-step gain is 1.93 points [1.14, 2.78] against a bar of 1.0) and is not described as saturating. The same-vendor curve meets it, at 0.23 points, but that route is sent temperature 0 and its verdict varies across readings on only 3 of 110 defective instances under this configuration, so its flatness is not evidence of exhausted headroom; whether it reflects the sampling or the model is not separated here (Section B.2).

reaches 32.7% [20.7, 45.0] at 10.7% [5.9, 16.1] with four readings, against the shipped route’s 30.0% at 16.0% with eight, but no paired contrast establishes both advantages together.

The rulebook. We added one sentence to the auditor’s rulebook: *find what the visible tests do not cover*. At eight readings on all 110 defective instances, it raised union recall from 30.0% to 60.9% [48.6, 72.5], +30.9 points [19.1, 43.1] (preregistered), at a price of 36.7% [28.5, 44.8] false positives against 16.0%. An earlier revision experiment asked whether the changed flags make code pass. There, at one reading on 56 defective instances (41 problems), flags rose from 10 to 25, +26.8 points [13.6, 40.4] (exploratory; exact McNemar $p = 0.0003$, cluster sign-flip $p = 0.0009$). On correct code they rose by 12.5 points, an interval [−3.6, +26.8] that includes zero. An improvement in outcomes was not established: on the contrast that isolates the rule, the hidden-suite pass rate after one revision moved +5.4 points [−0.9, +12.1]. The sentence changes what the auditor writes; whether that makes code pass is not established.

What is missed. Fifty-seven of the 110 defective instances were flagged by no reading of the first three families. Two model raters, one of them the agent that ran this programme, agree post hoc that in 44 of them (77.2% [62.1, 91.1]) the specification’s prose does not entail the value the hidden suite expects on the first failing input, as in the example that opens this paper. The category is broad: it covers prose

that is silent, prose that allows the program’s value, and prose that determines a value the suite contradicts. That reverses an earlier classification of the same instances as unexercised edges. Being undetermined is not enough to be missed: reading six families (five distinct models, one of them also read with the added rulebook sentence) at forty readings shrinks the never-flagged set to 32 of 110, and 43 of the 68 instances both raters call undetermined are flagged by some reading. The undetermined share of what remains is similar, 25 of 32 (78.1%), but the 32 are a subset of the 57, so this is a description, not a test of stability. Of twelve MBPP tasks an outside paper names as ambiguous, incomplete or contradictory (Richter & Papadakis, 2026), 11 are in our defect population, and our frozen labels call 9 of them undetermined (post hoc). An attempt to build a population whose specifications determine the defect by construction did not deliver that property, so this remains a post hoc description (Appendix B.5).

What follows. A union recall figure is contingent on the false-positive rate it cost, the model and sampling of the route, the rulebook, the decision rule, and what the benchmark’s specifications determine. Nine further studies, reported in Appendix B.5 with what each licenses, include a specification-rewriting intervention that raised adjudicated diagnosis only where the added rule did not wholly contradict the hidden suite, and a second substrate that closed without a quotable number.

Table 1. Selected quantities of the general-code studies (Section 3 and Appendix B), with the population each is measured on and the label its source report gives it. Values are percentages or percentage points. Intervals are 95% problem-cluster percentile bootstraps unless the label names another method. The coverage simulation of §A.3 measures single-rate intervals only; the contrasts and the category-conditioned shares have no measured coverage. The two third-rater rows rest on labels from a rater that agreed with itself on 7 of 11 repeated items (6 problems). The correct-code row counts disagreement on inputs extracted from the findings; of the 20 flagged instances, 9 agreed on every extracted input and 9 yielded no valid input, which can be neither confirmed nor refuted. Every row is read from a committed record by `figures/make_table1.py`.

quantity	population	value	interval	label
Union recall, shipped route, $K = 8$	110 defect, 56 problems	30.0%	[20.0, 40.7]	preregistered
Union false positives, shipped, $K = 8$	150 correct, 143 problems	16.0%	[10.1, 22.3]	preregistered
Pooled union, 3 families	110 defect	48.2%	[36.7, 60.0]	preregistered
its false positives	150 correct	36.0%	[28.2, 44.2]	derived
Gain from the eighth reading	110 defect	+1.93	[1.14, 2.78]	bar 1.0, not met
Same-vendor stronger — shipped, recall	110 defect	−26.4	[−37.3, −15.6]	preregistered
its false-positive difference	150 correct	−12.7	[−19.3, −6.1]	prereg.; 3.3% vs 16.0%
Rulebook rule — shipped, recall	110 defect	+30.9	[+19.1, +43.1]	preregistered
its false-positive cost	150 correct	+20.7	[+12.8, +28.8]	preregistered
Rulebook rule, flags	56 one-revision defect	+26.8	[+13.6, +40.4]	exploratory
its cost on correct code	56 one-revision correct	+12.5	[−3.6, +26.8]	unconditional, approx.
isolating contrast, pass rate	112 one-revision, 96 problems	+5.4	[−0.9, +12.1]	registered, failed
astra — shipped, common $K = 4$	110 defect, 56 problems	+11.5	[+2.9, +21.3]	exploratory
its false-positive difference	150 correct	−0.3	[−3.8, +3.4]	exploratory
Undetermined, sheet groups	121 entries, 110 instances	+19.8	[+1.2, +38.1]	registered method
one definition of caught	57 against 53	+23.3	[+2.0, +43.6]	post hoc
Undetermined share, 3-family residual	44 of 57	77.2%	[62.1, 91.1]	post hoc
6-family residual	25 of 32	78.1%	n/a	post hoc
Correct-code flags, reference disagrees	2 of 20 flagged	10.0%	[0.0, 26.3]	preregistered
Clarified — original, diagnosis	32 selected, 19 problems	+28.1	[+7.7, +51.4]	one-signed; exact $p = 0.0625$

4. Auditing AI for science

A scientist would hand an auditor three jobs: check the code that computes a result, check that a reported result follows from that code, and check the data an analysis starts from. We ran one preregistered study for each, with ground truth no model supplies: scientists’ numeric tests for code, re-execution for results, and injected faults for data. Every item is audited through the product’s own audit call and shipped rulebook, with the same flag rule as above. Construction, gates, amendments and full results are in Appendix C.

4.1. Scientific code

Design. SciCode (Tian et al., 2024) decomposes research problems from 16 subfields into steps with numeric tests written by scientists. The same frozen generator wrote every step three times, each sample building on its own earlier steps. An instance is correct if it passes the step’s tests and defective if not; a step enters only if some sample passes it, which excluded 184 of 336 evaluated steps. This gives 81 defective instances (30 problems) and 150 correct ones (53 problems). Labels are per step, but the auditor sees the whole chain, and 51 of the defective and 49 of the correct instances have a failing earlier step (post hoc description). SciCode has no visible/hidden split, so the auditor sees the step’s prose, background, header and code, and no tests. The shipped cross-vendor auditor and the

generator’s own model each read every instance eight times. After the registered arm’s outcomes, a second registered arm changed only the task text, naming the step’s function as the deliverable instead of opening with the whole problem; the program bytes are identical (Appendix C.1).

Results. What the task named as the deliverable changed the flag rate substantially (Figure 2a). With the whole problem as the task, the shipped auditor flagged 87.7% [72.7, 98.6] of defective and 68.7% [56.6, 79.7] of correct instances at eight readings, and some of its objections to correct code were that one step did not implement the whole problem. With the task reframed around the step, flags on correct code fell to 36.0% [26.4, 45.6], a paired change of −32.7 points [−42.8, −22.4]. Recall fell too, to 72.8% [59.3, 87.3], a change of −14.8 points [−28.6, −2.7], though the cluster sign-flip test for it gives $p = 0.0625$. The change is the whole reframing (the step named, the problem description moved last and marked as context, earlier functions exempted), and the arms ran in sequence. The generator’s own model’s union recall was 46.9 points lower than the shipped auditor’s in each arm ([−63.0, −30.7] and [−61.5, −34.4]; a route comparison), while flagging 27.3% [18.4, 37.3] and 18.0% [11.5, 25.6] of correct code, respectively.

At one reading in the reframed arm, the shipped auditor flags 61.7% [47.9, 75.9] of defective and 19.8% [12.9, 27.2] of correct instances: it discriminates, modestly. We read a

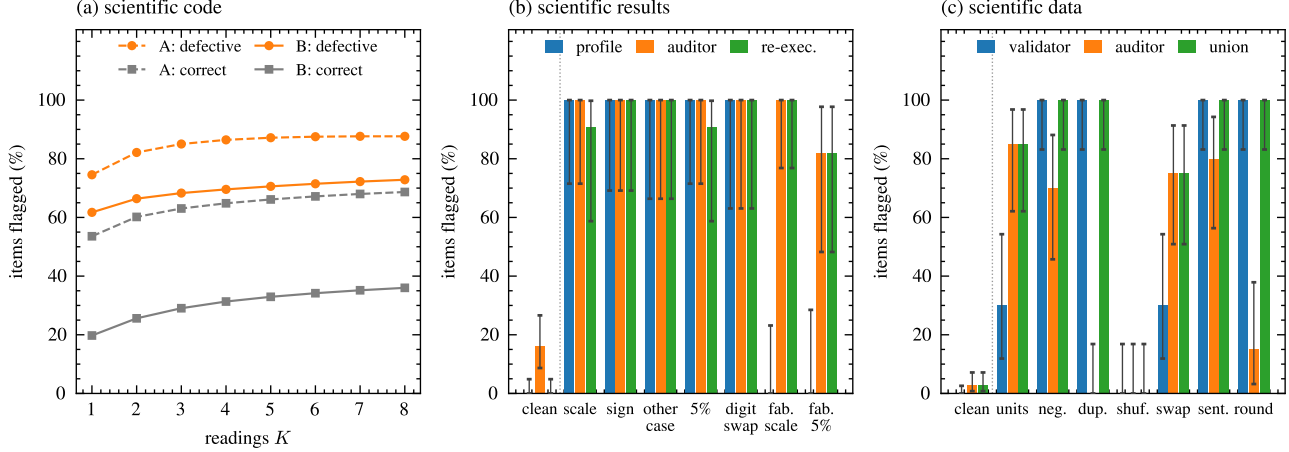


Figure 2. What each tier flags. (a) Scientific code: the shipped auditor’s union flag rate against the number of readings, on defective (orange) and correct (grey) instances, in arm A, whose task gave the whole problem (dashed), and arm B, whose task was reframed around the step (solid). (b) Scientific results and (c) scientific data, by fault type, with clean items leftmost: the deterministic tier (blue: the science profile in b, the documentation-derived validator in c), the model auditor at four readings (orange), and re-execution (b) or the union of the two (c) in green. In (c): mixed units, negated values, duplicated rows, shuffled target, swapped columns, `-999` sentinel, rounding. The model auditor’s flag is any model BLOCKER in the union of readings; the profile’s is a hard failure, the validator’s any fired check, re-execution’s a value outside `np.allclose` defaults. Bars carry exact binomial 95% intervals on the item counts (supplementary; they ignore clustering).

seeded sample of 20 flagged correct instances whose earlier steps all pass, taking the first BLOCKER of the first flagged reading of each; four of the twenty concern earlier functions that the reframed task excluded from the deliverable. They range from real divergences between code and step text, through contradictions inside the benchmark’s own instructions, to tests too weak to separate outputs; which flags point at defects was not established. An independent expert audit of SciCode found self-contradictory specifications, non-reproducible gold answers and over-tight tolerances that reject correct solutions (Hu et al., 2026), so some instances labelled defective here may be correct code the benchmark rejects. Of the 81 defective instances, 7 and 19 escaped all sixteen readings of the two families in the two arms. Both raters judged 4 of 7 (two problems; cluster interval $[0.0, 80.0]$; $\kappa = -0.235$) and 13 of 19 (eight problems; $[28.6, 93.5]$; $\kappa = 0.587$) undetermined by the prose, too few problems and too little agreement for a stable share. The registered test that this share exceeds the general-code residual’s 77.2% is killed in both arms by its own interval rule.

4.2. Scientific results

Design. Each of 74 correct SciCode instances (36 problems) whose output on one seeded test case is a real scalar or a short real array becomes a results increment: the program, the test’s inputs, a script that runs one on the other and writes a run log, a `results.json` citing each value’s log line, and a report whose numbers carry the product’s number→source annotations. Each yields a clean item and

a faulty one. Five fault types are *reporting* faults, where the artefacts disagree with each other or with the log; two are *fabrications*, a $\times 10^3$ slip or a 5% error written consistently into log, results and report. The model auditor reads each item four times with the shipped `science` profile on; the profile’s own verdict comes from the same call; and re-execution compares the output with the report (Appendix C.2).

Results. The deterministic profile flags all 49 reporting faults and none of the 25 fabrications, because in those everything agrees with everything else (Figure 2b). The model auditor flags 23 of the 25 fabrications, 21 with a sound finding about the fault (post hoc), usually by working the result out from the code, for example evaluating a KL divergence to 1 where the report says 1000; the other two rest on wrong calculations. All 23 discordant items go the model’s way (exact McNemar $p = 2.4 \times 10^{-7}$). On reporting faults its rate is of the combined workflow, since its prompt carried the profile’s findings. Re-execution misses four faults whose change to a near-zero output falls within `np.allclose`’s default absolute tolerance. Setting those four aside (post hoc), re-execution flags 70 of 70 faulty items at 0 of 70 clean, against the model auditor’s 68 of 70 at 12 of 70. On the registered population the two flag 70 and 72 of 74 faulty items, and eight of the model’s twelve clean-item flags identify real construction defects, so this subset favours re-execution for injected numerical faults without making it the better check in general. The model auditor also flags 12 of 74 clean items. Read one by one (post hoc), 8 correctly identify defects in our construction

(a provenance sentence false where the benchmark passes inputs as call literals; a unit label contradicting documented units); 4 rest on wrong calculations or a syntax error that is not there.

4.3. Scientific data

Design. Four CC BY 4.0 physical-science tables (Yeh, 1998; Brooks et al., 1989; Tüfekci, 2014; Hamidieh, 2018); 280 items of 120 rows with a data card giving each column’s meaning and unit: 140 clean, and 140 carrying one of seven faults (mixed units, negated values, duplicated rows, a shuffled target, swapped columns, a –999 sentinel, rounding). Both model families read each item four times, against a documentation-derived validator written from the card and frozen before any item existed (Appendix C.3).

Results. The two tiers flag overlapping but different faulty items (Figure 2c). The validator flags 92 of 140 faulty items (65.7% [57.9, 73.6]) and no clean item; the shipped auditor 65 of 140 (46.4% [38.6, 54.3]) at 4 of 140 clean items (2.9% [0.7, 5.7]); their union 112 (80.0% [73.6, 86.4]) at the same 4. These intervals resample items within dataset; four datasets are too few to resample, so no interval carries between-dataset variation. The auditor’s 4 clean flags and 4 of its faulty-item flags are requests for fuller column descriptions on one card; counting only findings about the injected fault (post hoc), the auditor flags 61 and the union 110. Of the 20 faulty items the auditor flags and the validator misses, 18 carry a finding about the injected fault itself (post hoc), mixed units or swapped columns; in one it argued that three atoms spanning a 135 pm radius range cannot have a mean radius of 1.6 pm, the injected cell being that value converted to ångströms. It flagged no shuffled target and no duplicated rows (0 of 20 each), and did worse than the validator on every fault the validator was written for. The generator’s own model flags 67 of 140 clean items, and for at least 48 of those (a post hoc lower bound) one finding states a row count, field count, missing column or truncation the file contradicts. The validator omits one range the concrete documentation states and uses a list of continuous columns the card does not give.

Held-out diagnosis at a fixed budget. The counts above rest on flags, and their diagnosis counts are post hoc. A further preregistered study asked the question a referee would: on 280 new items built the same way from a new seed, does the model auditor *correctly diagnose* faults the validator misses, at a clean-flag rate fixed in advance? Diagnosis was decided by code, not by a model: a flagging finding counts when it names the injected column (both columns for a swap; duplication for duplicated rows), read from the injection manifest. The rule is necessary, not sufficient, for a correct diagnosis. Each family’s operating point was fixed on the items above before the new ones existed: the

smallest vote threshold over four readings whose clean-flag rate was at most 10%, which is any one reading for the shipped auditor and all four for the generator’s own model. On the held-out items the validator diagnoses 90 of 140 faulty items (64.3% [56.4, 72.1]) at no clean flag, and the validator with the shipped auditor 105 (75.0% [67.6, 81.9]) at 9 of 140 clean flags (6.4% [2.8, 10.5]): +10.7 points [6.0, 15.9], 15 items one way and none the other (exact McNemar $p = 6.1 \times 10^{-5}$), the registered primary. At its own operating point the generator’s own model diagnoses 40 at 14 clean flags, 12.1 points [4.1, 20.3] fewer than the shipped auditor (a route comparison; Appendix C.3).

5. Discussion

A recall number is an operating point. Recall and false positives move together with any severity threshold, and a critic can be tuned to any recall by blocking more. On general-purpose code the same auditor sits at different points under different decision rules, repetition buys recall at a roughly constant price in false positives, and pooling families buys coverage at a higher one. On scientific code the operating point also moved with what the task named as the deliverable. A recall figure quoted without its false-positive rate, its rulebook, its decision rule and its task framing does not carry to another setting.

What the science studies add. The science profile’s design rule is that the model names evidence and code verifies it. The science studies give that rule an empirical reading and a boundary. On results, number→source flagged every inconsistency between a report, its results file and its log, and could not see a result that was wrong consistently; the model flagged 23 of 25 such fabrications, 21 with a sound finding about the fault (post hoc). On data, a validator written from the card flagged what a rule can state and missed what needs reading across columns and units; of the 28 mixed-unit or swapped-column items among the 48 faulty items the validator missed, the model flagged 18 with a finding about the fault (post hoc). Where both tiers ran, the deterministic one flagged no clean item, and every clean-item flag and every wrong finding came from the model. This is partly by construction: a clean results item was admitted only if the profile and re-execution passed it, and the clean data pools had exact duplicates and impossible rows removed before sampling. On data the union beat either part; on results the model’s flags already contained the profile’s, whose findings its prompt carried. Neither study shows that the model tier scales. The artefacts were small, the model’s fabrication findings are hand calculations (two of the 23 rest on wrong ones), and in these runs it flagged no duplicated rows and no shuffled target (0 of 20 each). A model reading is a useful second tier where the artefact fits in the reader’s working memory. It is not a

substitute for executing the code or computing over the data.

Clean items must be clean by the auditor’s standard. In the results and data studies the auditor flagged four defects in our own construction: a card and file that named a column differently, a report claiming a run no file performed, a provenance sentence false for inputs passed as literals, and a unit label contradicting the code’s documentation. Each was a correct flag on an item we had built as clean. Two surfaced before the reported readings and were repaired by amendment; the other two surfaced by reading every flag on a clean item, which the registrations required. A fifth construction defect, data files placed outside the audited scope, was not a model finding: it showed up because every verdict came back as an escalation. In the code study the task we wrote named the whole problem when one step was delivered, and a sample of the auditor’s flags on correct code objected to exactly that; there we read a sample, not every flag. Without such reading, a false-positive rate can measure the evaluator.

Limits. The auditor families are not sampled alike, which bounds every cross-family union contrast, and no family was compared at a matched false-positive rate. Primary intervals are problem-cluster bootstraps (items within dataset for the data study) whose coverage was simulated only for single rates. SciCode itself contains specifications that contradict their tests (Hu et al., 2026), so some of its defective labels may be wrong. Every rating was made by a model, one of them the agent that designed the studies; no human rated anything. The reviewer of every report is also one of the measured families. The code and results studies use one benchmark and the data study four tables, all with small artefacts and one generator; the results and data faults are injected. The full limits, and the experiments that would remove them, are in Appendices D and E.

6. Related work

Critics and repeated reading. Trained critic models catch more bugs in model-written code than unaided humans and hallucinate some (McAleese et al., 2024); aggregating several review passes raises review F1 on real pull requests (Zeng et al., 2025); and coverage under repeated sampling grows roughly log-linearly with the number of samples over four orders of magnitude (Brown et al., 2024). The natural-defect setting, the precision cost of a louder critic and the diminishing returns of repeated sampling are therefore known. What we add is narrow on this axis: the union of independent readings as an instrument for locating a limit, ground truth supplied by an interpreter, and the false-positive rate beside every recall figure.

Who audits whom. Self-correction without external feed-

back does not reliably improve reasoning (Huang et al., 2024), and model judges prefer their own generations, harmfully on instances they get wrong (Chen et al., 2025). Our same-vendor arms are the code-audit case of that literature; because routes differ in model, operating point and sampling, we report their contrasts as contrasts between routes, not as self-preference.

What the hidden suite decides. EvalPlus rebuilt HumanEval and MBPP with larger suites (Liu et al., 2023); models collapse onto one reading of an underspecified task, and twelve MBPP tasks were named as such by hand (Richter & Papadakis, 2026). Generated tests err mostly in the expected value (Xiong et al., 2023), and generated oracles tend to capture what a program does rather than what was intended (Konstantinou et al., 2024). Our claim is the auditing counterpart: where the prose does not fix the value, a recall number computed against that suite measures the benchmark as much as the auditor. Consensus filters for generated tests (Taherkhani et al., 2026; Xu et al., 2025) and self-consistent errors (Tan et al., 2025) bear on one of our appendix studies (Appendix F).

Auditing scientific work. On manuscripts, no model exceeds 21.1% recall or 6.1% precision on errata-grade errors and repeated runs rarely rediscover the same errors (Son et al., 2025); the best models detect 46.7% of real paper-code discrepancies (Baumgärtner & Gurevych, 2026). An LLM auditor of AI-scientist projects detects hand-labelled pitfalls far better with logs and code than with the paper alone (Luo et al., 2025); pattern-based linting targets methodology bugs in scientific Python (Samsonau, 2026); cleaning agents find row-level errors in corrupted tables and miss errors spanning rows (Bendinelli et al., 2025); and simulation code can run and still solve the wrong equations (Song et al., 2026). SciCode supplies scientist-written tests (Tian et al., 2024), and an expert audit of it found many specifications and tests that reject correct solutions (Hu et al., 2026); reproduction benchmarks ask agents to recover reported results (Siegel et al., 2024). On constructed statistical workflows, LLM reviewers detect planted flaws almost always but flag 17–50% of correct scripts (Chelate & Heuer, 2026), and Luo et al. (2025) already report both true- and false-positive rates. Scientific auditing and paired operating points are therefore not new. What we add is this measurement programme applied to scientific code, and the comparison of the model tier with deterministic verification on reported results and data, which we did not find in prior work.

Preregistration. Preregistration of evaluation protocols is beginning to appear in machine learning (Vaccaro, 2026; Thomas et al., 2026); we count our use of it as method, not contribution.

Impact Statement

This paper measures how well automated auditors of model-written code, results and data work, so its main consequence is for how much such auditors should be trusted. The measurements argue against two kinds of over-trust: a recall figure quoted without its false-positive rate and task framing, and a deterministic provenance check read as a guarantee that a result is right, when it verifies only that artefacts agree with each other. If adopted, the practices we argue for (reading every flag on items built as clean, reporting operating points, pairing a model reader with executed checks) raise the cost of audits but reduce false assurance in scientific and engineering work. A risk is that published limits are used to game an auditor, for example by fabricating results consistently; we report that weakness because a user of such systems needs to know it. All data are public benchmarks and datasets under permissive licences, and no human subjects were involved.

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A. Method in full

A.1. Substrate, population and ground truth

Every general-code result (Section 3 and Appendix B) is measured on one substrate: HumanEval+ and MBPP+ (EvalPlus). These are Python functions whose median task is 41 words of prose and 6 lines of reference code, each with an executable hidden test suite. An interpreter therefore decides every stratum, recall denominator and false-positive population, and no model judges a recall or false-positive outcome. The one outcome defined by adjudication, correct diagnosis in the clarification study, is judged jointly by two models and is labelled so where it appears. The benchmark’s base suite is shown to the generator as the *visible* suite, and its plus suite is held back as the *hidden* one. We also built and audited a second substrate, the standard-library half of BigCodeBench. It closed without a quotable number when its preregistered criterion was not met on a third run, and nothing from it is reported here.

How the population was built. The corpus has 542 problems. Two, HumanEval/32 and Mbpp/590, were excluded by name before generation because their own reference solutions fail their own hidden suites in this harness, which leaves 540. One frozen generator, `claude-haiku-4-5-20251001`, was shown each specification and its visible suite and wrote one solution per problem. A candidate that passes the visible suite and fails the hidden one is a *defective* instance; one that passes both we call *correct*; one that fails the visible suite is discarded from every recall and false-positive population. The defect stratum is small, about 10% of solutions. Under a rule registered before the second pass, and depending only on execution outcomes, we therefore repeated generation over the same 540 problems until there were at least 100 defective instances or three passes had run. It stopped at two passes: 56 and 54 defective, 455 and 455 correct, 29 and 31 discarded. An instance is therefore a (pass, problem) pair, and one problem can contribute two. Recall is measured on all 110 defective instances, which come from 56 problems. False positives are measured on 150 of the 910 correct instances, from 143 problems; that sample was drawn once from a fixed seed before any auditor ran. Seven of the 110 defective instances fail by not terminating rather than by a failed assertion, and the registered population keeps them.

Two consequences of these definitions hold throughout. First, *correct* means *passes the hidden suite*. It is a benchmark label produced by an interpreter, not a guarantee that the program meets its specification, and a suite can be incomplete or wrong. Since one of this paper’s results (§B.4) is that a specification often fails to determine what the hidden suite asks, we cannot treat passing that suite as correctness in the ordinary sense, and a *false positive* is likewise a flag on test-passing code rather than a demonstrated mistake by the auditor. Second, the defect population is not a sample of model-written code: it is the residue that already survived the visible tests, which is by construction the harder part. A recall figure here is the fraction of that residue an auditor catches, never the fraction of all defects a model writes.

A.2. The auditor and the estimand

The auditor reads the specification, the visible tests and the candidate, and returns findings with severities; an increment is blocked when any finding is graded BLOCKER. It never sees the hidden suite. The estimand throughout is union recall at K : the fraction of defective instances blocked by at least one of K independent readings, averaged exactly over all $\binom{K_{\max}}{K}$ subsets rather than by sampling. The matching quantity on the correct stratum is the union false-positive rate, and a union recall figure is reported only beside the false-positive rate it cost.

Five auditor routes appear below, with the short labels the figures use: the shipped cross-vendor auditor `gpt-5.6-terra` (cross); the same-vendor `claude-haiku-4-5` (self), the generator’s own family and size; the stronger same-vendor `claude-sonnet-4-6` (self-strong); `claude-opus-4-8` (self-frontier), read at $K = 4$; and `gpt-6-astra` at high reasoning (astra), read at $K = 4$. The auditor’s constitution, its prompt builder and every route’s capability card are in the auditor harness, a separate repository, at the commit named in the provenance file, whose per-reading records carry the hash of every prompt actually sent.

The routes are not sampled alike, and this bounds every cross-family union-of- K contrast in the paper. The cross-vendor route’s capability card does not permit a sampling temperature, so no sampling parameter is sent and the route varies between readings. The same-vendor routes permit one, and the layer sends temperature 0, so those routes are nearly deterministic. A union over readings that do not differ cannot grow, so the two families’ unions are not equally free to rise with K , and a cross-family difference in union recall confounds auditing ability with how much each union could grow. The asymmetry is a property of the shipped routing layer we measured rather than a design choice. We did not discover it until after the runs, and we report the $K = 1$ contrast beside the $K = 8$ contrast wherever it matters. The asymmetry is the first item in §D.6.

A.3. Statistics

The unit of observation is the instance and the resampling unit is the problem, because problems recur across generation batches and instances sharing a problem are not independent. The K readings of an instance are repeated measurements of it, not K further independent observations. The bootstrap over problems estimates uncertainty from which tasks the benchmark happens to contain. It does not estimate how much a fresh batch of model calls on the same tasks would move the numbers, which would need replicate runs we did not budget for at every arm.

Every rate carries a 95% percentile bootstrap interval over whole problem clusters (10,000 resamples, seeds fixed in each preregistration). Where the source report gives a Wilson interval, it appears beside the bootstrap, marked as too narrow. Paired contrasts carry an exact McNemar test and a cluster-level sign-flip permutation test. Where every discordant pair points one way, the percentile bootstrap's bound at zero is an artefact, and an unconditional interval (grid-approximated, unclustered) is quoted instead. The one exception is the clarification study, whose registration fixed the bootstrap for a contrast that turned out one-signed; its interval is reported as registered and read beside its exact cluster test (§B.5). A saturation curve is summarised by a constrained exponential fit, and the fitted asymptote is quoted only when the preregistered flattening bar is met; otherwise it is labelled an extrapolation and the raw union at $K_{\max}$ is the number quoted. Counts of five or fewer are quoted as counts. Each study handles multiplicity as it registered. The study carrying the primary contrasts corrects over every comparison it computed: a Bonferroni family of sixteen with a threshold of 0.00313, which two comparisons clear. Other studies registered a single primary, and the second-vendor rulebook study reports its twenty further quantities without correction.

We measured interval coverage by simulation rather than assuming it. The simulation uses the two strata's actual cluster profiles (110 instances in 56 problems, and 150 in 143) and a within-problem concordance of 0.5, over 400 replicates of 600 resamples each. The table gives the percentile bootstrap's coverage of a nominal 95% rate interval. The Monte Carlo standard error is about 0.011 near 95% coverage and 0.020 at the lowest cell. The simulation covers single rates only, and it uses integer-index percentiles where the analyses interpolate (`analysis/coverage-simulation.py`).

true rate	0.03	0.05	0.10	0.16	0.30	0.48	0.60
defect (56)	0.798	0.870	0.933	0.933	0.953	0.935	0.928
correct (143)	0.888	0.918	0.938	0.933	0.943	0.955	0.965

In these simulations coverage is near nominal at the true rates this paper's headlines sit at and badly anticonservative at small true rates: at a true rate of 0.03 on the defect stratum the simulated interval covers about 0.80, not 0.95. This is evidence of potential undercoverage for analogous intervals, not a measured coverage of any interval quoted here. Intervals on rates below roughly 0.10 may be narrower than they claim. The simulation costs no model calls. Two paired-contrast constructions tried in this programme covered 0.416 and 0.075 where 0.95 was claimed, and neither is used. No sample size was justified in advance by a minimum-detectable-effect calculation.

An interval that contains zero is reported as one that does not exclude zero, and nothing more. No equivalence margin was registered anywhere in this programme and no equivalence test was performed, so none of those intervals is evidence that two conditions are alike or that an intervention did nothing. Where a hypothesis fails its registered criterion we say it failed, not that the effect is absent.

All analysis of the general-code studies is plain Python 3.13 using only the standard library (`random`, `math`, `itertools`, `csv` and `json`). The bootstrap, the exact McNemar test, the cluster sign-flip permutation and the exact subset enumeration behind every union-at- K figure are implemented in the repository rather than called from a numerical package, so each can be read where it is used and none depends on a third-party version.

A.4. Preregistration and independent review

Who did what. The studies were designed, run, analysed and written up by an AI research agent, Claude (Anthropic; `claude-opus-5` and `claude-fable-5-1`, working in Claude Code), under the direction of the author, who is responsible for the work. Wherever a rating or adjudication below is attributed to the agent, it was that agent. It wrote the rubrics, knew the hypotheses and the earlier labels, and is from the same vendor as the generator. No human rated or adjudicated anything in this paper. Every report was then reviewed by a second model, `gpt-6-astra`, as described below.

Each study that made model calls was preregistered before its first one, fixing its population, hypotheses, primary outcome,

intervals, kill conditions, budget and stopping rule, with two exceptions. First, the severity sweep of §B.2 re-grades findings already archived, makes no model calls, and was never registered: every figure it produces is post hoc and exploratory, and none is a confirmatory test. Second, the analysis behind §B.5’s third-rating result was registered a day *after* its contrast had been computed and committed, in a document that claimed the opposite. That document is withdrawn in the repository, and the analysis is registered nowhere. Its method was fixed in advance by an earlier amendment, but the disjoint reading, the abstention-excluded secondary and the broken-sheet comparison were not.

Planned departures are numbered amendments, each recording what was known at the time, and in the general-code studies each was committed before the step it governs, with one exception: Amendment 1 of the study that added the `self-strong` and `self-frontier` routes was committed after 30 of its second-draw readings had landed, not before them as it first stated, and that study’s Amendment 2 corrects this on the record. In the science studies two were not: A4S-3’s Amendment 1 was committed together with the rebuilt items it governs, and A4S-2’s Amendment 2 after the two pilot readings it already governed. Both are recorded as errata in the registrations (Appendix G). Deviations discovered only afterwards, such as a leak check or a spend guard implemented more narrowly than registered, are disclosed in the report and are not amendments. Where an amendment followed an interim observation, the results say so and argue whether the change could favour the hypothesis.

Each report was then reviewed by a model from a vendor other than the generator’s. The reviewer had read access to the raw records and was asked to re-run the analysis, recompute the intervals independently, and check every sentence against the records. That reviewer is `gpt-6-astra`, which is also one of the auditor families measured in §B.2, one of the two raters behind §B.4’s labels, and one of the two adjudicators whose joint labels define the clarification study’s outcome. For most claims this is a cross-vendor check on work the reviewer had no part in. For the claims that rank `astra` against the other families, and for the clarification study’s adjudication, the reviewer is the subject, and those results should be read as self-assessed until a model that audited nothing in this programme has checked them. For the same reason, the clarification study’s Amendment 10 disqualified this model as the study’s manipulation-check rater. It kept the model as that study’s second adjudicator and its reviewer, however, so the standard was applied unevenly within one study and not applied to the review.

Reviews are archived verbatim, including those that refused approval, and several studies needed many rounds. On one day five reports were refused at once, and clearing those five took thirty-one further rounds. In most of those rounds, what changed was what the reports claimed about their measurements. Most of the later refusals objected to a repair written to satisfy the previous round, not to the study it repaired. A fix written to satisfy a reviewer is itself an unreviewed claim.

The reviewer is a language model, and this paper measures the ceiling on language models reading work they did not produce. That applies to our own instrument. The studies of §B.5 cleared review at rounds 3, 3, 4, 6, 7, 9, 10 and 11, and the ceiling study behind the headlines at 21. In every case the round before the last still returned substantive findings. Some of those findings were defects introduced by the repairs between rounds, so the counts do not estimate a single pass’s recall; they show only that no report cleared in fewer than three. *Quotable at round N* means a report survived *N* passes, not that it was verified, and the marks in this paper should be read as a filter of unknown recall.

Two further points follow. First, measurements did move on review. The severity sweep’s first review found its loader misreading one family’s draws and another family’s population, and every cross-family figure was recomputed. The clarification study’s second review found an instance wrongly excluded, which moved its primary from 9 of 31 to 9 of 32 instances and its contrast from +29.0 to +28.1 points. And a label count inside a reviewed report was 71 where its data say 67. Once a figure’s inputs were right, the reviewers reproduced rates, intervals, subset enumerations and seeds independently and found no arithmetic error, and most refusals were about what the prose claimed. The reviews were reliable on recomputation and unreliable on interpretation, and the two should not be quoted as one assurance. Second, the class of defect this leaves open is the one that looks right to a language model. The clarification study audited the wrong program for 384 readings, because a per-problem field was used where a per-instance one was needed. Every automated gate passed. Each checked that the code was *identical across conditions*, and it was, because every condition showed the same wrong program. The agent found the error while reading an adjudication sheet, by noticing that a displayed program returned exactly what its hidden suite was said to reject. Neither the gates nor a reviewer found it, and we do not know what else of that kind remains.

Every review prompt in this programme stated the claimed result before the reviewer began. We tried to test what that costs, and the attempt failed. One report, quotable at round 6 after five refusals and unchanged since, was dispatched again under a prompt whose own text withheld every figure and every prior verdict and asked for an independent computation first.

That prompt’s list of binding artefacts then named two files that state those figures, one of them the claims-ledger entry for the result under review. The blinding was not enforced, so the agreement between that pass and the briefed one establishes nothing. What the pass did return is a disagreement count: six substantive findings that six briefed rounds on the same file had not produced. Among them, the report’s two groups were two different definitions of *caught* rather than one population with duplicate rows; its rating sheet displays at most three failing inputs while 81 of its 121 entries have more recorded; and a count of changed labels was 71 where the data say 67. The disagreement count is $n = 1$, uncontrolled and unregistered. A seventh round might have found new things under any prompt, and no matched briefed round was run against the same file. We report it as a reason to test blinding properly rather than as a result about blinding. Later first rounds withheld the conclusions from the prompt text, but the report under review stayed among the named artefacts, with an instruction to derive the result before reading it, and that request does not blind the reviewer.

Every study drawn on here has completed that review. One further study was re-run twice and closed without a quotable number; the paper names it only to say what it no longer supports.

B. Results on general-purpose code in full

The claims this paper defends, each with its interval and its label, are listed in `CLAIMS.md`. A claim enters that file only after its report has passed an independent review by a model from a different vendor with access to the raw records, and every section below draws only on such reports. Section B.5 reports eight studies whose reviews took between three and eleven rounds each to reach that point, and states what each does and does not license. The headline results of §B.1–§B.4 come from the ceiling study; where they cite a study from §B.5, they say so.

Rates carry the 95% problem-cluster percentile bootstrap as the primary interval. Problems recur across generation batches, and an interval that treats instances as independent is too narrow. Where the source report gives a Wilson interval, it appears beside the bootstrap. For single rates, simulations of an analogous interval under this clustered design (§A.3) suggest coverage near nominal at the headlines’ rates and possible undercoverage below a true rate of about 0.10, under an assumed within-problem correlation. They do not measure the coverage of the intervals reported here, and the paired contrasts and category-conditioned shares have no measured coverage.

Sixteen comparisons were computed in the study that carries the primary contrasts, under one Bonferroni family with a threshold of 0.00313. That correction is applied to p values and never to intervals. Every interval in this paper is an unadjusted 95% interval, so a directed statement resting on an interval alone, with no p value that clears its family, is worded as a direction rather than as a finding. Each p value belongs to the family of the study that computed it. Those in §B.3 are members of that sixteen-member family. The one in §B.2 is the auditor-family study’s single registered primary, so there was no family to correct within; it would also clear 0.00313. The clarification study’s p values are reported beside a primary registered on its interval, and no correction was registered for them. Where a p value appears below, the exact McNemar value and the cluster-level sign-flip permutation value are both given, because instances within a problem are not independent and only the second respects that.

Table 1 (Section 3) collects selected primary quantities with their populations and labels; the prose gives each one where it is used.

Table 2 lists the general-code studies before their numbers appear: what each read, what it changed, what it measured and how its main quantity was registered.

B.1. Returns to repeated reading diminish short of complete

The shipped cross-vendor auditor reads an increment holistically and returns findings; an increment is blocked if any finding is graded BLOCKER. Over eight independent readings of the same 110 defective instances, union recall rises from 10.7% at one reading to 30.0% [20.0, 40.7] at eight (Figure 1). The eighth reading still adds 1.93 points, cluster [1.14, 2.78], against the flattening bar of 1.0 points registered before the first model call, so the bar was not met. Under exact subset averaging that statistic has a closed form: the number of instances flagged by exactly one of the K readings divided by Kn , here $17/(8 \cdot 110)$. It is a count of singletons, not a fitted slope. Its interval resamples problems and so carries task-sampling variance. The statistic itself is made of reading-sampling variance, which the interval does not describe. The bar should therefore be read as a statement about these 17 instances, not about a population slope.

The constrained fit puts the asymptote at 31.5%, but on our own registered criterion that figure is an extrapolation, and the

What moves the ceiling on automated audit, from code to science

Table 2. Study map, general-purpose code. Every study reads the same frozen generator’s candidates (110 defective instances from 56 problems, 150 correct from 143) unless the population column says otherwise. “Route comparison” marks contrasts that change model and sampling together.

Study	Population	What varies	Main quantity	Status
Repeated reading (§B.1)	110 defective, 150 correct	readings $K = 1-8$; shipped cross-vendor vs same-vendor route	union BLOCKER recall and false positives	preregistered primary
Stronger routes (§B.2)	as above	two stronger same-vendor models: self-strong at $K = 8$, self-frontier at $K = 4$	union recall of self-strong at $K = 8$	preregistered primary; route comparison
Decision-rule sweep (§B.5)	archived findings of the two studies above	four grading rules; no model calls	union recall under each rule	not registered
Rulebook revision (§B.3)	56 defective (41 problems) and 56 correct, one reading	one added rulebook sentence; one revision round	flags on the 56 defective; hidden-suite pass after revision on all 112	outcome contrasts registered; flag contrasts exploratory
Referent rule, second family (§B.5)	as repeated reading, $K = 4$	the same sentence, and a grading rule, on self-strong	union BLOCKER recall	preregistered
Referent rule at $K = 8$ (§B.5)	as repeated reading	the same sentence on the shipped auditor	union recall at $K = 8$	preregistered
Residual rating (§B.4)	57 instances no reading of three families flagged	rubric: does the prose entail the expected value	share rated undetermined	rating registered; association post hoc
Outside rater (§B.5)	121 residual and flagged entries	a rater outside the residual’s families	share undetermined, missed vs caught	not registered (a named exception)
Specification edit (§B.5)	32 selected residual instances	added rule vs same-length placebo vs original, $K = 4$	jointly adjudicated diagnosis	preregistered
Test validation (§B.5)	frozen generated tests and candidates, including candidates failing their visible tests	two corroboration rules	share of retained failing test applications that are wrong	preregistered; criterion not met
Injected defects (§B.5)	constructed defects and unmodified twins	injected edit	blocks, $K = 8$	descriptive only
Flags on correct code (§B.5)	150 correct, $K = 8$	finding texts kept	share of flagged instances that disagree with the reference on an extracted input	preregistered
Second substrate (§B.5)	BigCodeBench standard-library half	substrate	—	closed; nothing reported

raw union at eight readings is the number to quote. What the data support is diminishing returns at the budget we could afford, not saturation: at 1.9 points per reading the curve was still climbing when we stopped. The bar also depends on where a family was stopped. Exact subset-averaged increments do not grow with K (this curve’s is 2.9 points at the fourth reading and 1.9 at the eighth), so a fixed bar is harder to meet at $K = 4$ than at $K = 8$, and meeting it at either says little by itself about where the curve ends.

Twenty readings pooled across three auditor families reach 48.2% [36.7, 60.0], and the same twenty readings flag 54 of 150 correct increments, 36.0% [28.2, 44.2]. That is more than twice the shipped route’s 16.0%, so pooling buys its coverage at its own price. Those twenty readings are also not twenty opportunities. The pool is eight cross-vendor, eight same-vendor and four *astra* readings, and the same-vendor route gains 1.8 points from its first reading to its eighth against the cross-vendor route’s 19.3. It is sent temperature 0 and returns nearly the same verdict each time, though this design cannot say how much of that is the temperature and how much the model. Counting that route as roughly one reading puts the depth nearer thirteen than twenty (a heuristic, not an estimate), and the pooled figure should be read as a union over three routes rather than over twenty independent looks.

The price is measured on the same footing. Union false positives on 150 correct increments reach 16.0% [10.1, 22.3] at eight readings. We also compute the recall bought per point of false positive: recall gained over one reading divided by false positives gained over one reading. It is 1.68 at eight readings and changes little as readings accumulate. From $K = 2$ to

$K = 8$ the ratio runs 1.67, 1.63, 1.64, 1.65, 1.66, 1.67, 1.68. It falls once, at $K = 3$, and then rises, so repeated reading on this route buys recall at a slightly improving price. The ratio is a quotient of two measured rates and carries no interval of its own.

B.2. Which model reads, and under which severity rule

Replacing the auditor with a stronger model of the generator’s own vendor moved union recall at eight readings by -26.4 points, cluster $[-37.3, -15.6]$: 4 of 110 against 33 of 110. Three instances were flagged only by the new family and thirty-two only by the shipped auditor, at an exact McNemar p of 4.2×10^{-7} and a cluster sign-flip p of 2.0×10^{-5} . This was the auditor-family study’s single preregistered primary, so no family correction applies. Three measurements qualify what it means, and none of them favours reading it as a statement about auditing ability.

First, the two auditors are not at the same operating point, and this paper elsewhere warns against comparing auditors at different operating points. The stronger model flags correct code at 3.3%, or 5 of 150, which §A.3 says to quote as a count. Its bootstrap interval, $[0.7, 6.7]$, sits in the range where simulations of an analogous interval cover about 0.89 at a true rate of 0.03 rather than 0.95, so it may be narrower than it claims. The shipped auditor flags 16.0% $[10.1, 22.3]$. The stronger model blocks less of everything.

Second, the sign reverses under a weaker severity rule (Figure 3). That rule counts any finding rather than only findings graded BLOCKER; it is exploratory and was not preregistered. Under it the stronger model reaches 59.1% $[47.3, 70.6]$ on the defect population at 34.0% $[26.5, 41.7]$ on correct code, and the shipped auditor reaches 34.5% $[23.6, 45.5]$ at 19.3% $[12.8, 26.3]$. The ratio of recall to false positives, a derived point ratio with no interval, is 1.74 for the stronger model and 1.79 for the shipped auditor, so nothing here shows one family buying more recall per false positive than the other. Under any-finding grading the stronger model’s findings cover more instances, and under the BLOCKER rule fewer, so the -26.4 depends on the counting rule. That does not show why the families’ findings differ, or that any finding identifies the hidden failure.

Third, the two families are not sampled alike. The shipped cross-vendor route’s capability card does not permit a temperature, so no sampling parameter is sent and the route varies between readings; the same-vendor route permits one, the layer sends temperature 0, and that route is nearly deterministic. A union over nearly identical readings cannot grow much, and it does not: over $K = 1$ to 8 the cross-vendor family gains 19.3 points and the stronger same-vendor family gains 3.0. At one reading the gap between them is -10.1 points, not -26.4 . The rest accumulates as the two unions grow at different rates, and because model and sampling differ together, the design cannot say how much of that difference is the sampling and how much the model. We regard this as the most serious limitation of the contrast, and §D.6 records what would be needed to remove it.

A third family points the other way and belongs beside the headline. A high-reasoning model of the cross-vendor family reaches 32.7% $[20.7, 45.0]$ union recall at four readings while flagging 10.7% $[5.9, 16.1]$ of correct code, against the shipped auditor’s 30.0% at 16.0% with eight. Paired on the same instances, the two advantages are never established together. At a common four readings it leads on recall, $+11.5$ $[2.9, 21.3]$, with an unresolved false-positive difference of -0.3 $[-3.8, +3.4]$; against the shipped auditor’s full eight it leads on false positives, -5.3 $[-10.3, -0.7]$ unadjusted, but not on recall, $+2.7$ $[-6.5, +13.0]$ (§B.5). The evidence therefore does not support “a stronger model is a worse auditor”. It supports the narrower statement that this particular same-vendor model, under the preregistered BLOCKER rule and sampled at temperature 0, blocks far less.

A same-vendor route of the generator’s own size reached lower union recall than the cross-vendor one on these instances, -12.7 points $[-25.0, -0.9]$ at eight readings, signed so that a negative value favours the cross-vendor route. The interval is unadjusted, its upper end is 0.9 points from zero, and that route is sent temperature 0, so this is a direction under this sampling and not a ranking of auditors. A study on a second substrate was meant to test whether the sign reverses there; it closed without a quotable number, and on its clean run the same contrast included zero, so no reversal was established either (§B.5). The figure above is a measurement on this substrate, and whether it is a property of same-vendor auditing is open.

B.3. One rulebook sentence moves flags, and does not measurably move outcomes

Adding a single rule to the auditor’s constitution, telling it to find what the visible tests do not cover, moved flags on the defect population from 10 of 56 to 25 of 56: $+26.8$ points, cluster $[13.6, 40.4]$, exact McNemar $p = 0.0003$, cluster sign-flip $p = 0.0009$. Together with the pooled defect-and-correct flag contrast, this is one of only two comparisons that clear the

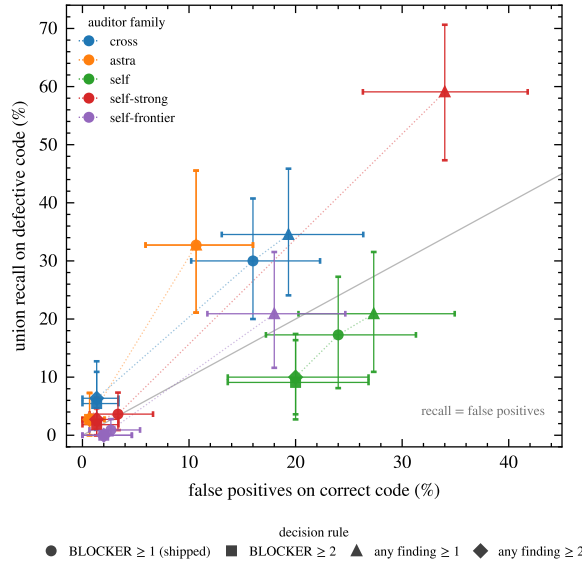


Figure 3. Union recall is a property of the decision rule, not only of the auditor. Each point is one auditor family re-graded under one of four decision rules on the same archived readings, with problem-cluster 95% intervals on both axes; no new model calls were made. The families are read at different depths: cross, self and self-strong at all eight of their readings, astra and self-frontier at their four. The figure therefore places each family at its full archived depth and is not an equal-depth comparison; the common-depth contrasts are in Table 1. The dotted guides join a family’s four points and are not a measured curve. Two things Table 1 cannot show: the families do not lie on one frontier, and self-strong has no point under the four deterministic rules between 3.3% and 34.0% false positives, exactly where a comparison matched to the cross-vendor auditor would sit, so its ranking there is bracketed rather than measured. Nothing in this figure assesses finding content; a flag is not a diagnosis.

Bonferroni threshold, and the two are the same effect seen twice.

The estimand is narrower than that of the sections above and must be read at its own scale. It is measured on 56 instances from 41 problems, not the 110-instance defect population. It is recall at a single reading, union recall at $K = 1$ on that population, because every loop arm revises once. And it is exploratory, because the preregistration named outcome contrasts rather than flag contrasts. For those reasons we do not compare its magnitude with the -26.4 or the $+19.3$ of the preceding sections, which are union-recall differences at $K = 8$ over 110 instances (a contrast between routes and a gain from $K = 1$).

Two costs travel with it. On correct code the same rule moved flags by $+12.50$ points. Every discordant pair points the same way (7 against 0), so by the rule stated in §A.3 the percentile bootstrap’s bound at zero is an artefact, and the admissible interval is the unconditional one. That interval is $[-3.6, +26.8]$ and does not exclude zero. It is approximate rather than guaranteed exact, because it uses a grid-approximated nuisance supremum and ignores clustering. Its exact McNemar and cluster sign-flip p are both 0.0156, against that study’s Bonferroni threshold of 0.00313. The directed statement the records support is the pooled one, where 23 discordant pairs point one way and 1 the other (exact McNemar $p = 3.0 \times 10^{-6}$, cluster sign-flip $p = 1.0 \times 10^{-5}$): the sentence makes the auditor louder overall. On the correct stratum alone the cost is $+12.50$ points with an interval that contains zero.

The second cost is to outcomes. The contrast that isolates the rule compares the arm with the added rule against the cross-vendor arm on hidden-suite pass rate after one revision, over both strata (112 instances on 96 problems). The effect is $+5.36$ points, cluster $[-0.89, +12.07]$, exact McNemar $p = 0.146$ and cluster sign-flip $p = 0.183$; the interval does not exclude zero. The rule demonstrably moves what the auditor writes down. Whether that converts into code that passes the hidden suite is not established, and since this paper argues that a flag is not a catch, that distinction applies to our own intervention first.

B.4. Most of what the auditor missed is failure the specification never determined

Fifty-seven of the 110 defective instances were flagged by no reading of the three families read at that point. That is a statement about twenty readings, not about detectability. If an instance is caught with probability p on each independent reading, twenty readings miss it with probability $(1 - p)^{20}$, which is still 98% at $p = 0.001$ and 36% at $p = 0.05$. Never

flagged here does not mean never flaggable, and we use the residual as a population to characterise rather than as a set of defects shown to be invisible.

We can put a number on how flaggable it is. Reading six families at forty readings (five distinct models), including the same auditor under a rulebook with one sentence added, shrinks the never-flagged set from 57 to 32 of 110, and 43 of the 68 instances both raters call specification-undetermined are flagged by some reading of some family. Being undetermined is not sufficient for being missed. What does not move is the composition: the undetermined share of the never-flagged set is 44 of 57, 77.2%, at three families and twenty readings, and 25 of 32, 78.1%, at six families and forty readings.

The first classification of that residual asked whether the visible tests exercised the failing input before it asked anything else, and it called 46 of 57 unexercised edges. Re-rating the same instances under a second protocol reverses that. The second protocol asks first whether the specification’s prose determines the hidden suite’s expected value, redefines the categories by entailment, and uses two raters in consensus; ordering, definitions and raters all changed together, and the study cannot isolate which mattered. The two raters agree that 44 of 57 (77.2%, cluster [62.1, 91.1]) are cases where the prose does not settle the value, and that 2 of 57 are cases where it does and the visible tests merely never built the input. The preregistered condition for withdrawing “the residual is mostly unexercised edge” fires on both of its clauses, and that claim is withdrawn. The interval is conditional on a category and has no committed coverage simulation, so it is uncalibrated, as is every category-conditioned interval we report.

A classification that reverses when its protocol changes is, by itself, weak evidence about the auditor, and the raters are not independent of it: one was the agent that ran this programme (§A.4), which wrote the rubric and knew the earlier labels and which claim was at stake. Three things support the second reading. The rubric was fixed before any label was written. The labels were committed before we looked for outside evidence. And when a first pass was found to have carried instance identifiers and a sheet name into the raters’ view, the labelling was re-run with identities stripped and the result held. An outside party had also independently recorded specifications as defective, in twelve MBPP tasks named in a paper written for another purpose (Richter & Papadakis, 2026). Of the 11 that fall inside our defect population, our raters agree on 9, and both disagreements lie on the boundary that paper’s own “incomplete” category draws. That check is one-sided by construction: it can support agreement where an outsider looked, and can bound nothing about the rate.

A rater outside the residual’s families, asked a different question. One of the two raters above is gpt-6-astra, whose own misses help define the residual. A model that is not one of the families defining it, gpt-5.6-luna (which read these instances only as a secondary audit route in an earlier architecture study), later rated the same 57 instances post hoc under a different rubric: whether the prose determines an answer at all, on a sheet showing at most three failing inputs and never the expected values (§B.5). It labels 38 of 57 undetermined, 13 determined and 6 cannot-tell. The two figures measure different constructs. The earlier category also admits prose that determines an answer the oracle contradicts, which this rubric does not count as undetermined, so 38 is not a corrected estimate of 44. What they share is that a participating and a non-participating rater both put a majority of the residual on the undetermined side. The non-participating rater agreed with itself on only 7 of 11 repeated items.

B.5. Nine further studies, and what they license

Eight further studies cleared independent review after between three and eleven rounds each, and a ninth, on a second substrate, closed without a quotable number. We report them here rather than folding them into the sections above because several are narrow or negative results whose limits are part of the claim, and because the shape of their refusals is itself a finding about this method.

A rulebook sentence moves a second vendor’s family. The referent rule, *find what the visible tests do not cover*, raises union BLOCKER recall on self-strong (claude-sonnet-4-6) at four readings by +20.9 points, cluster [10.8, 31.8], at a false-positive cost on correct code of +8.0 points [3.4, 12.7]. This reproduces §B.3’s effect on a different vendor’s model at a different K , with its own price. A second intervention, a rule on how findings are graded, shows no demonstrated improvement: +0.9 points [−1.8, +4.5]. That hypothesis fails its registered criterion, and the failure demonstrates neither equivalence nor that the rule did nothing. The same study also measured, post hoc, whether a flagged finding *reports a failure* on the instance’s failing input class, that is, says that class raises or returns the wrong value. The count is 3 of 110 (2.7% [0.0, 7.3]) under the shipped constitution and 27 of 110 (24.5% [14.4, 35.7]) under the referent rule (not the grading rule), and it may not be quoted as defect recognition. Its rubric admits a finding even where it grades the class non-blocking or says the specification is silent, so findings that disclaim the requirement are counted in.

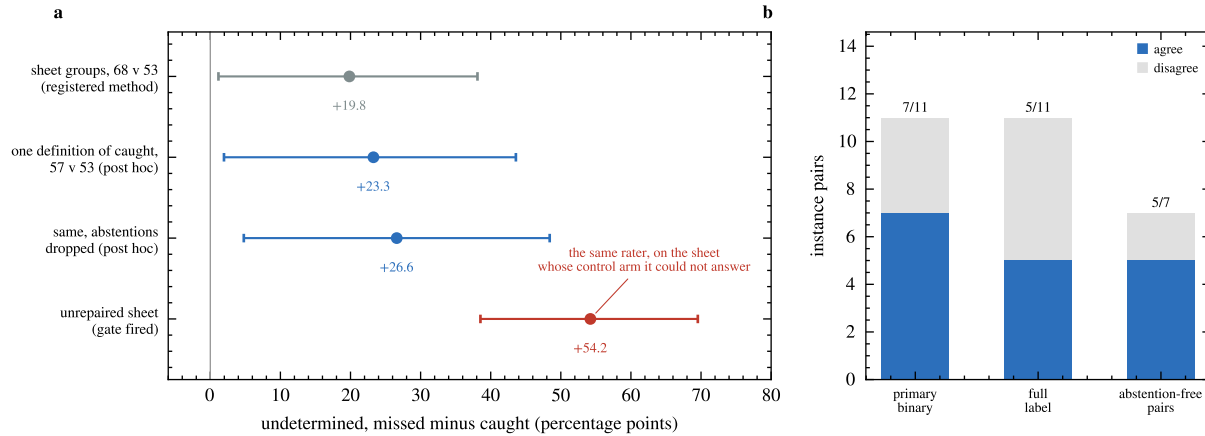


Figure 4. The residual contrast, printed beside the instrument that produced it. a, The share of instances a rater judged specification-undetermined, missed group minus caught group, under each reading, with problem-cluster 95% intervals. The two sheet groups use different definitions of *caught* and share eleven instances, so the two readings that apply one definition are the ones that compare populations; both are post hoc. The red point is the same rater on the same sheet before its control arm was rebuilt, when 42 of 53 caught entries carried no failing-input evidence and all 42 drew *cannot-tell*. With that arm the contrast was larger and pointed the way the hypothesis points. The repair and a fresh model run changed together, so presentation, sampling and batch context are not separated, and a preregistered one-third *cannot-tell* gate excluded the earlier figure. b, The same rater's agreement with itself on the eleven instances the sheet carries twice, with identical text and identical evidence. The contrast in a is read off the instrument in b, and the claim this paper draws from it is stated under that constraint.

The added rule raises union recall to 60.9%; a higher ceiling is not established. On the defect population the referent rule takes union recall from 33 of 110 to 67 of 110, or 60.9% [48.6, 72.5]. This is the same quantity as the shipped auditor's eight-reading headline in §B.1: blocked by at least one BLOCKER, unioned over eight readings, here for the shipped auditor with the referent rule against the shipped auditor alone. The gain is +30.9 points [19.1, 43.1], at a cost of +20.7 points [12.8, 28.8] in false positives. The single-draw false-positive rate is 23.6% [17.1, 30.3] and fails the product bar on all eight draws. The caution that a flag is not a catch attaches to both figures equally. What separates them is their price, 36.7% [28.5, 44.8] false positives against 16.0%, which is why the shipped route is the shipped route. The fitted asymptotes differ by +27.1 points [10.9, 39.0], and *that does not establish a higher ceiling*. The cross-vendor arm never flattened, so one of the two terms is an extrapolation. Heterogeneous low-probability detection imitates a ceiling at this budget, since an instance found with probability 0.02 per reading survives eight readings about 85% of the time. And this study's own zero-inflated estimator finds no never-detectable class in *either* arm. The study's adjudication kill is arithmetic rather than a labelling artefact. Only 9 of 110 instances drew any finding on the single reading whose finding texts were archived, so the largest defect-asserting count available under perfect adjudication was 9, below the floor of 20. Every adjudication-derived figure in the study is conditional on labels whose production can no longer be audited. One rater's raw reply, prompt, launcher and execution log lived only in a session scratchpad, and they were destroyed between the review that flagged them and the repair.

Corroboration does not validate a generated test. A wrong generated test, one that fails on the dataset's canonical solution, is not reliably recognised by asking whether further independent generation draws also fail on the same candidate. Neither rule that makes further generation calls reaches the preregistered threshold on the registered population. That population covers failing applications on all three candidate strata; 58 of its 107 failing applications are on candidates that already fail their visible tests. The retained wrong-test rates are 12.2% [3.6, 22.6] (Wilson [7.0, 20.6]) for majority-over-draws and 14.9% [6.5, 25.5] for any-draw agreement, against a limit of 2%. Neither is shown to improve on the free within-draw comparator at 12.8% [3.9, 23.9]: the better of the two rules keeps the same wrong applications plus four correct ones, a small observed advantage the study cannot resolve. A post hoc analysis restricted to the candidates a product would meet, those passing their visible tests, gives 6 of 33, 18.2% [0.0, 44.8], and 10 of 43, 23.3% [6.8, 47.1]. The first leaves its relation to the 2% threshold unresolved; the second's unadjusted interval lies wholly above it, which is exploratory evidence of exceeding it. Neither is a preregistered test on that population, whose registered kill rule was a point estimate on the mixed population. A floor of 2.4% holds on the original unique-test quantity because 29 of the 45 wrong tests never fail on any candidate at all. The claim is negative and bound to this substrate: these rules do not validate generated tests here, which is not to say generated tests cannot be validated.

An injected population does not give the residual result a prospective test. We constructed a population of injected defects intended to make specification-determinedness hold by construction. It does not. The filter registered to recover each edit’s first failing hidden input is a truthiness check on a field the injector wrote: of 281 recorded witnesses, 179 are strings and 102 are lists, and none is executed. Four of the remaining five filters accept cases their registered wording excludes, such as a suite that reports success after an early exit, a process that terminates without an assertion or an exception, and an added import from a module already imported. No filter establishes specification entailment, and repairing the first one alone would not have: recovering the failing input and executing a witness shows that the edit changes behaviour on an input the hidden suite exercises, not that the specification entails the value expected there. What may be quoted is descriptive. At eight readings the auditor blocked 90 of 92 accepted injected instances and 11 of their 92 unmodified twins, with 79 discordant pairs one way and none the other. On the natural residual it blocked 33 of 110. The instances the construction’s own gate *refused* were blocked 31 of 40, which is why that gate cannot be assumed conservative. A probe separates the injected set from the natural one at 96.7% on the items it answered, *by a feature it does not identify*: the two arms share no problem, so task composition alone could produce the result, and the probe’s model is one of the two gate models. §B.4 therefore keeps exactly the status its own study gave it, post hoc and unreplicated. A successor needs an independent party to validate the actual violation *and* the specification-entailing expected behaviour under a fixed, blinded protocol. Neither a second substrate nor an independent rater suffices alone, because task composition and construction confounding survive both.

A study on a second substrate closed without a quotable number. In a second benchmark, the visible-test text shown to the auditor *and* to the generator was sliced out of its test class without the class header, so it failed to parse on all 300 tasks while scoring executed the intact class. The repair for that fault introduced a second one, re-attaching a hidden method’s decorators to the first visible method on 26 tasks. A third run on a clean extraction reproduced five of the six reversals of first-substrate findings that the second run had produced, and did not reproduce the sixth. The criterion registered before those numbers existed was stated over the set, so the strict reading applies: the study is closed after three runs and about \$82, with no quotable number. Whether five of six should have licensed the five is recorded as an open question in CLAIMS.md.

Measured union recall is rule-dependent, and a headline’s magnitude belongs to its own rule. We re-graded findings already archived by two of the studies above under four decision rules, with no new model calls. The shipped BLOCKER threshold’s estimate stays intact for its own rule, which *was* preregistered as “ ≥ 1 BLOCKER finding”, and *reverses its sign* under any-finding grading. That reversal is a mechanical consequence of changing what counts, and it identifies nothing about why the families produced different findings or different severities. At a common depth of four readings, *astra* exceeds the cross-vendor auditor on union recall over the defect stratum by +11.5 points [2.9, 21.3], paired within instance, with a false-positive difference of -0.3 points $[-3.8, +3.4]$. The two sides are not averaged alike. The cross-vendor arm has eight readings and is averaged over all $\binom{8}{4} = 70$ four-draw subsets, while *astra* has exactly four, so $\binom{4}{4} = 1$ and its estimate is a single realisation with none of its reading-selection variance averaged out; one arm’s luck in four draws is fully in the number and the other’s is not. On the cross-vendor auditor’s complete eight-draw ladder the same contrast is +2.7 points $[-6.5, +13.0]$ with a false-positive difference of -5.3 points $[-10.3, -0.7]$. That comparison points toward lower false positives, not higher recall. The false-positive interval’s upper end is 0.7 points from zero, and it is not adjusted for the other contrasts computed beside it, so this is a direction the data lean and not one they establish. Observed false-positive rates near one another are not a matched operating point. One family has no point under the four deterministic rules between 3.3% and 34.0% false positives, precisely where a matched comparison would sit, so its ranking is bracketed rather than measured. Nothing here assesses finding content against the hidden failure, so none of it speaks to what any finding identified.

The ambiguity gap survives a rater that did not write the rubric, and the instrument that shows it disagrees with itself. The contrast in §B.4 between missed and caught instances had been rated by two models, one of them the agent that wrote the rubric. A third model re-rated the whole sheet after we rebuilt it on mechanically recovered witnesses. That rebuild was necessary because the control arm had been unanswerable: 42 of 53 caught entries carried no failing-input evidence, and all 42 drew *cannot-tell*. The share labelled *undetermined* is 43 of 68 against 23 of 53, +19.8 points [1.2, 38.1]. Those two groups are not two populations: they define *caught* differently, and 11 instances fall in both. Removing those 11 from the missed group gives +23.3 points [2.0, 43.6], post hoc and larger, so the overlap had understated the contrast. The limit is part of the result. On those 11 instances, shown with identical text and identical evidence, the rater agreed with itself on 7 of 11 on this binary outcome and 5 of 11 on the full label, where the earlier study’s two raters each agreed 11 of 11 (Figure 4). Every interval’s lower end lies within five points of zero, and this remains a model; no human has rated this material. The *unrepaired* sheet scored +54.2 points [38.5, 69.5], 2.7 times the repaired contrast and in the direction that flatters the hypothesis. That is a descriptive association between one repair and one fresh re-rating, not evidence about specifications: the repair, the run and its sampling changed together, and the preregistered one-third *cannot-tell* gate

catches it.

Adding the missing rule to the specification moves diagnosis, but only where the rule did not wholly contradict the oracle. The contrast above conditions on the specification; this study changes it. The population is the 44 never-flagged instances that both earlier raters called specification-undetermined. For the 32 on 19 problems that passed the generation gates, a model shown the failing *inputs*, but never their expected values, wrote one behavioural rule into the prose, and an auditor re-read the same code and the same visible tests under the original, the clarified and a same-length placebo specification, four readings each. An instance counts as diagnosed only if both adjudicators judged some finding to state the behaviour the hidden suite expects. Diagnosis rose from 0 of 32 to 9 of 32 under clarification and stayed at 0 of 32 under placebo. The registered primary, clarified against original, is +28.1 points, problem-cluster 95% interval [7.7, 51.4], which meets the registered criterion. The registered secondary uses the 23 instances where the re-rating study’s consensus (§B.4) and the third rater’s label agree; it is 8 of 23 against 0 of 23, +34.8 [10.0, 60.9].

Given the observed data, that criterion added little. Every discordant pair points one way, so no bootstrap resample can fall below zero, and with successes on five of 19 problems a resample is exactly zero only when it misses all five, with probability $(14/19)^{19} \approx 0.003$; the registered run has 27 such resamples in 10,000. Once five success-bearing problems are observed, the interval’s exclusion of zero adds little to them (with one such problem it would not have excluded zero), and §A.3 says to read a one-signed contrast from an exact test instead. The registration kept the bootstrap and we report it as registered, but the informative figure is the cluster sign-flip test the registration prefers. The nine successes sit on five problems, and the sign-flip test gives $p = 0.060$ (exactly $2/2^5 = 0.0625$ on enumeration), against 0.0039 for a McNemar test that counts nine independent pieces of evidence where five exist. It is not significant at 0.05. The registered naming-rate secondary was never computed.

Two further limits are part of the result. *The inferred rule contradicts the hidden suite entirely on 23 of the 32 instances and partly on 2*, and diagnosis was 0 of 23 where it contradicted against 9 of 9 elsewhere. That split was classified after the outcome was seen, not assigned, and the agreeing instances may simply be the easier problems; it does not say which property of the edits moved diagnosis. A separate model rated the clarified prose more determining than the original by +28.1 points [2.6, 54.3], but more determining than the placebo by +18.8 [−9.4, +45.2], so whether the placebo was inert as a specification is unresolved. One adjudicator is the agent that ran the study and the other is *gpt-6-astra*, the model that also reviewed this study (§A.4); they agreed on 84 of 91 items, $\kappa = 0.85$, which is agreement and not accuracy. The study supports a specification-edit contrast on these selected instances. It does not establish that underdetermination caused the original misses, and §B.4 is not strengthened into that claim by it.

What a flag on correct code points at. The false-positive rates above count flags on code that passes its hidden suite. A suite may leave behaviour unexercised, so a flag there need not be a false alarm. A preregistered study re-read the 150 correct instances with the shipped route at eight readings, keeping the finding texts the ceiling study had discarded; 20 were flagged (13.3% [7.9, 19.2], a fresh sample of the route’s 16.0%). A model given each finding, but not the reference solution or the hidden tests, extracted the concrete inputs it names, and candidate and reference were run on them. Two of the 20 disagreed with the reference on an extracted input the hidden suite does not exercise (10.0%, problem-cluster [0.0, 26.3]); on nine they agreed, and nine yielded no valid input. Because the interval includes zero, the registered reading is that at this resolution there is no evidence that this route’s flags on test-passing code identify reference disagreements. No class licenses the words “true positive” or “false alarm”: a disagreement does not show that the specification was violated, and agreement on an extracted input does not show a finding wrong.

What the refusals were about. These eight studies cleared review at rounds 3, 3, 4, 6, 7, 9, 10 and 11, fifty-three rounds in all. Most of those rounds ended in a refusal of the *repair* rather than of the measurement. Measurements moved on review in three of the eight: the sweep’s first review found its loader reading one family’s draws 4 to 8 as 1 to 5 and another’s population from a partial draw, and every cross-family figure was recomputed; a label count in the third rating’s report was 71 where its data say 67; and the clarification study’s primary moved by one instance and three classification rows (§A.4). Every figure above reproduced independently at every round after its last repair. The recurring failure was a check or a sentence that claimed more than it delivered. A binding protected one sentence of a withdrawal spread across six; when it was rewritten to protect all six, two of them were truncated before the words that carried their meaning. One check stripped HTML comments and called the result what a reader sees, and one test was named for a code path that never called it. A second pattern appeared in the last thirteen rounds: most of those refusals were *overstatements away from our own interest*, such as a registration whose chronology its own repository contradicted, a withdrawal that overcorrected past what the record supported, a publication prohibition a registered gate never imposed, and a limitation credited to another study’s

work that the other study had not performed. Understatement was corrected as often as overstatement; in both cases the fix was a smaller claim written where the claim was.

B.6. What follows for a recall number

Taken together, the results above say an audit recall figure is a joint property of the auditor and its material. It moves with which route reads (model and sampling together, which this design does not separate), with what the rulebook tells it to look for, and with the severity threshold at which a finding becomes a block, which is rarely reported. What it misses is associated with whether the hidden suite asks a question the specification answers. That association is observed in the residual and in a specification-edit contrast on selected instances; it is not isolated as a cause.

C. The science studies in full

The general-code studies (Section 3) measured the model tier on short general-purpose functions. A scientist would hand an auditor three different jobs: check the code that computes a result, check that a reported result follows from that code, and check the data an analysis starts from. We ran one preregistered study per job, each with ground truth that no model supplies. Scientific code (§4.1) is judged by scientists’ numeric tests, results (§4.2) by re-execution, and data (§4.3) by the faults we injected. Every item in all three is audited through the product’s own audit call with its shipped rulebook, and the same flag rule holds throughout: an item is flagged when the model grades any finding BLOCKER.

C.1. Scientific code

Substrate. SciCode (Tian et al., 2024) decomposes 80 research problems from 16 subfields into 338 steps, each a function with numeric tests written by scientists. Its targets are released separately from the code, which lets the tests act as a hidden oracle. On the benchmark’s development split, the gold code passes 48 of 50 steps in our environment; the two steps whose gold fails here are excluded, as are the three the official harness skips.

Population. The generator of the general-code studies wrote every step in SciCode’s own protocol, three samples per step, each sample building on its own earlier steps. A candidate is *correct* if it passes the step’s tests and *defective* if it fails or times out. A step enters either stratum only if at least one sample passes it, so a failure counts only where this environment has shown the step can be passed; 184 of 336 evaluated steps had no passing sample and were excluded. This yields 81 defective instances from 30 problems and 374 correct ones, from which 150 (53 problems) were drawn by seed. Labels are per step, but the program each auditor sees holds the sample’s functions for every step up to the one tested, and 51 of the 81 defective and 49 of the 150 correct instances have a failing earlier step in their chain (post hoc description).

What differs from the general-code studies. SciCode has no visible and hidden split, and we did not invent one. The auditor sees the problem, the step, its background, the function header, the allowed dependencies and the candidate, and no tests of any kind. The shipped cross-vendor auditor and the generator’s own model each read every instance eight times. We reach the generator’s model through a command-line route rather than the API. The route adds a short fixed context and uses default sampling, so this study’s same-vendor family is not temperature-0, unlike the one in the general-code studies.

Two arms. After the registered arm’s outcomes, a second registered arm changed only the task text, naming the step’s function as the deliverable; the program bytes are identical.

Residual. Defective instances flagged by no reading of either family are rated on a blind sheet under the general-code residual’s question (§B.4) adapted to science: do the step’s prose and background determine the value the test expects, including tolerance, units, conventions and numerical method. The raters are the agent that ran the study and a model that audits nothing here.

Results. In these runs, how the task framed the deliverable changed the flag rate substantially (Figure 2a). With the whole problem as the task, the shipped auditor flagged 87.7% [72.7, 98.6] of defective and 68.7% [56.6, 79.7] of correct instances at eight readings; some of its objections to correct code were that one step did not implement the whole problem. With the task reframed around the step, flags on correct code fell to 36.0% [26.4, 45.6], a paired change of -32.7 points [-42.8 , -22.4]. Recall fell too, to 72.8% [59.3, 87.3], a change of -14.8 points [-28.6 , -2.7], though the cluster sign-flip test for that recall change gives $p = 0.0625$. The change is the whole reframing (the step named, the problem description moved last and marked as context, earlier functions exempted), and the arms ran in sequence. The recall of the generator’s own model was 46.9 points below the shipped auditor’s in each arm ($[-63.0, -30.7]$ and $[-61.5, -34.4]$; a route comparison).

It flagged 27.3% [18.4, 37.3] and 18.0% [11.5, 25.6] of correct code in the two arms, respectively. At one reading in the reframed arm the shipped auditor flags 61.7% [47.9, 75.9] of defective against 19.8% [12.9, 27.2] of correct instances: it discriminates, modestly. We read a seeded sample of 20 flagged correct instances whose earlier steps all pass, taking the first BLOCKER of the first flagged reading of each; four of the twenty concern earlier functions that the reframed task excluded from the deliverable. They range from real divergences between code and step text, through contradictions inside the benchmark’s own instructions, to tests too weak to separate outputs; which flags point at defects was not established. An independent expert audit of SciCode’s test problems found self-contradictory specifications, non-reproducible gold answers and over-tight tolerances that reject correct solutions (Hu et al., 2026), so some instances labelled defective here may be correct code the benchmark rejects. Of the 81 defective instances, 7 and 19 escaped all sixteen readings of the two families in the two arms. Both raters judged 4 of 7 (two problems; cluster interval [0.0, 80.0]; $\kappa = -0.235$) and 13 of 19 (eight problems; [28.6, 93.5]; $\kappa = 0.587$) undetermined by the prose, too few problems and too little agreement for a stable share. The registered test that this share exceeds the general-code residual’s 77.2% is killed in both arms by its own interval rule. The rating sheets showed only the step’s function, not helpers its test calls, and truncated long expected arrays; of six items that appeared on both sheets, the second rater changed three labels.

C.2. Scientific results

Items. Each correct SciCode instance whose output on one seeded test case is a real scalar or a real array of at most ten elements becomes a small results increment: the program, the test’s inputs, a script that runs one on the other and writes a run log, a `results.json` citing each value’s log line, metadata declaring every input at a revision, and a one-paragraph report whose numbers carry the product’s number→source annotations. Seventy-four of the 150 correct instances qualify, from 36 problems. Each yields one clean item and one faulty item. The faulty item carries one of seven faults, assigned in rotation by seed. Five are *reporting* faults, where the artefacts disagree with each other or with the log the code wrote: a $\times 10^3$ scale slip, a sign flip, a value from another test case, a 5% relative error, and a transposed digit pair in the report alone. Two are *fabrications*, where the scale slip or the 5% error is written consistently into the log, the results and the report, so that every artefact agrees with every other and none agrees with the code.

Three auditors. The LLM auditor reads the increment four times, with the project’s checks set to the shipped `science` profile. The model’s prompt includes the deterministic result, as it does in the product, so on reporting faults the model has seen the deterministic finding. The deterministic profile’s own verdict comes from the same call. Re-execution runs the program on the inputs and compares the output with the reported values.

Gate. Before any model call, every clean item had to pass the science profile with no BLOCKER and re-execute to its reported value; all 74 did when first built. A two-reading pilot then found an instrument defect: no file in the item performed the run the report described, and the auditor blocked on that, correctly. We added the run script, discarded the pilot readings, and added a third condition to the gate, that each clean item’s run script reproduce its run log byte for byte. All 74 passed all three conditions with no repair.

Results. The deterministic profile does what it was built for (Figure 2b). It flags all 49 reporting faults, because the log, the results and the report no longer agree and number→source says where, and none of the 25 fabrications, because in those everything agrees with everything else. It flags no clean item. The LLM auditor flags 23 of the 25 fabrications at four readings, 21 of them with a sound finding about the fault (post hoc). Those findings work the result out from the code and inputs, for example evaluating a KL divergence of two small distributions to 1 where the report says 1000. The other two rest on calculations that are themselves wrong. Paired on the faulty items, all 23 discordant items go the LLM’s way (exact McNemar $p = 2.4 \times 10^{-7}$). A problem-level sign-flip test, run post hoc, gives 3.8×10^{-6} . On the reporting faults its rate is of the combined workflow, since its prompt carried the profile’s findings. Re-execution flags every fault except four whose change to a near-zero output falls within `np.allclose`’s default absolute tolerance; two of those outputs are floating-point zeros. Setting those four aside (post hoc), re-execution flags 70 of 70 faulty items at 0 of 70 clean, against the model auditor’s 68 of 70 at 12 of 70. On the registered population the two flag 70 and 72 of 74 faulty items, and eight of the model’s twelve clean-item flags identify real construction defects, so this subset favours re-execution for injected numerical faults without making it the better check in general.

The LLM also flags 12 of the 74 clean items. We read each of these flags (post hoc), and 8 are right. In 9 instances the benchmark passes the test’s inputs inside the call itself, so the report’s sentence that the program ran “on the inputs in `inputs.py`” was false; 7 of those clean items were flagged for it. Of four outputs whose unit the solution documents, two were flagged for our `dimensionless` label; one of the two is also among the seven. The remaining 4 flags rest on

wrong calculations or, in one case, on a syntax error that is not there. On this evidence a provenance chain and a reader that recomputes are complements: the first is exact about agreement and blind to consistent error, and the second sees consistent error but is sometimes wrong itself.

C.3. Scientific data

Substrate. We use four physical-science tables from the UCI repository, each under CC BY 4.0: concrete compressive strength (Yeh, 1998), airfoil self-noise (Brooks et al., 1989), combined-cycle power-plant output (Tüfekci, 2014), and superconductor critical temperature (Hamidieh, 2018). Exact duplicate rows and physically impossible rows (relative humidity above 100%) were removed from each pool before any item was drawn.

Items. Each item is 120 rows sampled from one pool with a data card that gives every column’s meaning and unit and, where the source states one, its range. There are 35 clean items per dataset and five per dataset for each of seven faults: mixed units in 20% of one column’s rows, three negated values of a non-negative quantity, 12 duplicated rows, the target shuffled among 30% of rows, two columns with disjoint ranges swapped, a -999 sentinel in 5% of one column’s cells, and a continuous column rounded to integers.

Auditors. Both model families read each item four times, with the dataset as the deliverable and the card as its documentation. The comparison is a deterministic validator written from the card alone and frozen before any item was built: declared ranges, non-negativity, duplicates, sentinels and integer-valued continuous columns. By construction it catches the negations, duplicates, sentinels and rounding. It sees mixed units or swapped columns only when they push a value below zero or out of a declared range, or make a continuous column integer-valued, and a shuffled target not at all. The question is whether the model catches what the validator cannot. Two instrument defects were found and fixed by amendment before the readings reported here: the card and file named one column differently, and the files first sat outside the audited scope, so the product had told the model there was nothing to audit.

Results. The two tiers flag overlapping but different faulty items (Figure 2c). The validator flags 92 of 140 faulty items (65.7% [57.9, 73.6]) and no clean item. The shipped auditor flags 65 of 140 (46.4% [38.6, 54.3]) at 4 of 140 clean items (2.9% [0.7, 5.7]), and their union flags 112 of 140 (80.0% [73.6, 86.4]) at the same 4. These intervals resample items within dataset; four datasets are too few to resample, so no interval here carries between-dataset variation. Of the 20 faulty items the auditor flags and the validator misses, 18 carry a finding about the injected fault itself, 9 of them mixed units and 9 swapped columns (post hoc). In one, the auditor reasoned that three atoms spanning a 135 pm radius range cannot have a mean radius of 1.6 pm, which is the cell the fault had converted to ångströms. The other 2, and all 4 flags on clean items, are requests for fuller column descriptions on one dataset’s card; so are 2 more of its faulty-item flags, which the validator also caught. Counting only findings about the injected fault (post hoc), the auditor flags 61 of 140 and the union 110, at no clean item. The auditor flagged no shuffled target and no duplicated rows (0 of 20 each), and it did worse than the validator on every fault the validator was written to catch. The generator’s own model, read through its command-line route, is not a usable data auditor here: it flags 67 of 140 clean items, and in at least 48 of those one finding states a row count, field count, missing column or truncation that the file contradicts (post hoc lower bound).

The deterministic validator is not neutral. It was frozen from the card before any item existed, but it omits one range the concrete source’s documentation states and uses a list of continuous columns the card does not give; with the omitted range it would flag 93 of 140. Unit conversion also left floating-point tails that mark some converted cells in two of the tables. None of the auditor’s mixed-unit findings cites them, but that does not show they had no influence.

Held-out diagnosis at a fixed false-positive budget. A fourth preregistered study (registered before any of its items was built or read) answers a question the study above cannot: whether the model auditor *correctly diagnoses* faults, rather than flags items, at a false-positive budget fixed before the data exist. It builds 280 new items by the construction above, unchanged except for the seed (no CSV is shared with the development items), and reads each four times with both families. Diagnosis is decided by code from the injection manifest: a faulty item is diagnosed when a flagging reading’s BLOCKER text names the injected column, both columns for a swap, or a duplication term for duplicated rows, using a frozen alias table. No model judges any outcome. The rule is necessary rather than sufficient for a correct diagnosis. On the development items it counts 60 of the shipped auditor’s diagnoses against the post hoc fault-relevant count of 61, and within one item on every fault type but swapped columns (13 against 15). The operating points were fixed on the development items: the smallest vote threshold over four readings whose clean-flag rate was at most 10%, any one reading for the shipped auditor (2.9% on development) and all four for the generator’s own model (7.1%); no threshold of the latter reaches 5%, which is why the budget is 10%.

On the held-out items the validator diagnoses 90 of 140 faulty items (64.3% [56.4, 72.1]) and flags no clean item. The shipped auditor at its operating point diagnoses 57 (40.7% [32.6, 48.9]) at 9 of 140 clean flags (6.4% [2.8, 10.5]), and the generator’s own model 40 (28.6% [21.7, 35.9]) at 14 (10.0% [5.6, 14.8]). The validator with the shipped auditor diagnoses 105 (75.0% [67.6, 81.9]) at the same 9 clean flags. The registered primary, the union against the validator alone, is +10.7 points [6.0, 15.9]: 15 items diagnosed only by the union and none only by the validator (exact McNemar $p = 6.1 \times 10^{-5}$), inside the budget, so the hypothesis holds. The gain comes from mixed units (5 to 13 of 20) and swapped columns (5 to 12 of 20), as on development. The secondary route comparison puts the shipped auditor 12.1 points [4.1, 20.3] above the generator’s own model (27 against 10 discordant items); model and sampling change together, and the two sit at different realised false-positive rates inside the same budget. Intervals resample items within dataset. The study cost \$24.41.

Read after the outcome (post hoc): of the six faulty items the shipped auditor flags without the rule counting a diagnosis, three are the card-documentation request below, one objects to the file’s row count, and two name one of the two swapped columns and describe its impossible values, which a reader would count and the rule does not. A seeded sample of ten counted findings each names the injected column and describes the injected corruption. All nine clean flags are on airfoil items and make one objection: the card names each column’s unit but does not define it, the same objection as the four development clean flags. The held-out items are new draws from the same four tables and seven synthetic faults, so the result does not extend to other data, faults or cards.

D. Discussion in full

D.1. What a union-recall number means

A union-recall figure depends on the false-positive rate it cost, so recall quoted without that rate cannot be carried to another setting. We make this as a methodological point. The measured support we tried to give it came from a second substrate that was run three times and closed without a quotable number (§B.5). The point needs no second substrate: recall and false positives move together with any severity threshold, and a critic can be tuned to any recall by blocking more. Our records support a narrower version on the substrate we report. The same auditor at two severity rules sits at two different operating points (§B.2), and the shipped auditor’s 30.0% [20.0, 40.7] recall is quoted only beside the 16.0% [10.1, 22.3] it cost.

The cleaner experiment for the general claim is a severity sweep within one substrate. Our sweep re-grades archived findings at four decision rules and shows each family’s recall and false-positive rate moving with the rule (Figure 3). Its rules are deterministic and do not place the families at a common operating point. A randomised mixture of two archived rules could reach one in expectation without new model calls, but it would be a new post hoc policy needing its own uncertainty analysis, and we have not built it. Two substrates differ in specification length, solution length, problem population, defect distribution and suite construction all at once, so comparing them could never have isolated the operating point.

D.2. Repeated reading buys nothing where the verdict does not vary

Union-of- K is the instrument this paper uses to look for a ceiling, and part of what it cannot measure follows from arithmetic alone. A union over readings that do not differ cannot grow, so the curve is flat for any route that returns the same verdict however often it is asked.

Our records establish the design fact that makes this matter here. The same-vendor routes are sent temperature 0, while the cross-vendor route is sent no sampling parameter (§B.2). From $K = 1$ to 8 the first family’s union gains 3.0 points and the second’s gains 19.3. No temperature-matched arm was run, so we cannot say how much of that gap is sampling and how much is the operating point. We treat it as a confound on our own contrasts.

For best-of- K , union-of- K or panel aggregation over critics, what follows is the arithmetic: a route whose verdict does not vary between readings gains nothing by being read again. We measured per-instance flag counts, not any distance to a decision boundary, and ran no panel aggregation, so we offer no law for how much a varying route gains. The diminishing shape of the curves is not itself evidence of a ceiling either. Exact subset averaging makes each reading’s marginal gain no larger than the last by construction, and the curves measure the size of those gains.

D.3. What the ceiling is made of

We do not rank these contributions, because they were measured on different populations, depths and rules. Two of them share a scale. Eight readings of one auditor buy 19.3 points of union recall. Twenty readings across three families buy a

pooled 48.2% [36.7, 60.0] at 36.0% false positives on correct code, and leave the complement, 51.8% [40.0, 63.3] of the defect population, untouched. Changing which model reads moves the same quantity, union recall at $K = 8$ over the same 110 instances, by -26.4 points $[-37.3, -15.6]$. That is a larger magnitude than repetition's 19.3, and the stronger route flags 3.3% [0.7, 6.7] of correct code against 16.0%. A same-size same-vendor substitution moved it by only -12.7 $[-25.0, -0.9]$, a direction under temperature-0 sampling. Neither number should be read as auditing ability, for three reasons given in §B.2.

Changing one sentence of the rulebook moves flags by 26.8 points [13.6, 40.4], and moves flags on correct code by $+12.5$ points $[-3.6, +26.8]$. This is an exploratory contrast on 56 instances at one reading, with no established effect on hidden-suite outcomes, and it is not on the scale of the other two.

A fourth contribution is not the auditor's at all, and we can describe it only post hoc. Under a rule that asks whether the specification's prose determines the hidden suite's expected value, 44 of the 57 never-flagged defects fail. We tried to build a population where the prose does determine the answer. Review established that no filter in it enforces that property, so it licenses description only (§B.5).

The association between never being flagged and never being determined is empirical. Selecting the missed instances does not fix their labels, and against caught instances the third rater's post hoc contrast is $+23.3$ points [2.0, 43.6]. That rater agreed with itself on only 7 of 11 repeated items, and the interval is conditional on its recorded labels. Selection and confounding rule out a causal reading, so the contrast locates the misses and does not estimate how much the benchmark contributes.

One intervention changes the specification instead of conditioning on it (§B.5). Adding a behavioural rule to the prose moved diagnosis on 9 of 32 such instances: six whose added rule agreed with the hidden suite, one where it broadened a precondition, and two where it partly contradicted the suite. Diagnosis moved on none of the 23 where the rule contradicted the suite entirely. That split was classified after the outcome was seen, not assigned. This is the only evidence here that manipulates the specification. Its population was selected for having been missed and called undetermined, and it cannot separate a specification that settles the question from one that settles it correctly.

The constructed population has two further limitations, one measured and one found in review. A frontier model separates the constructed edits from natural ones at 96.7% on the items it answered, and between 79.9% and 97.3% once the items it would not classify are counted at either extreme. It does so by a feature it does not identify. The two arms share no problem, and the model is one of the two that gated the population. And because the construction does not enforce specification-determinedness, it licenses *no* bound on the measured ceiling.

D.4. Flagging is not naming

An audit that blocks an increment for the wrong reason has still blocked it, and a recall number counts that as a catch. For one reading of the shipped auditor the finding texts were kept and adjudicated blind by two raters. The auditor returned a finding on 9 of 110 defective instances and asserted the actual failing behaviour on 5 to 7 of them, depending on how the two raters are combined. The raters agreed at $\kappa = 0.391$. We report that beside the rate because weak agreement is part of what the rate is worth. These figures are conditional on labels whose production can no longer be audited: one rater's raw replies and prompts were destroyed before the repair (§B.5). That study's preregistered kill condition fired on this result, and §B.5 states its headline in the form the kill requires. The figure describes one reading. It is not a naming rate for the eight-reading union, which would need the texts of all eight readings, and those were not archived.

D.5. What the science studies add

The science profile's design rule is that the model names evidence and code verifies it. The science studies give that rule an empirical reading and a boundary. On results, number→source flagged every inconsistency between a report, its results file and its log, and could not see a result that was wrong consistently; the model flagged 23 of 25 such fabrications, 21 with a sound finding about the fault (post hoc). On data, a validator written from the card flagged what a rule can state and missed what needs reading across columns and units; of the 28 mixed-unit or swapped-column items among the 48 faulty items the validator missed, the model flagged 18 with a finding about the fault (post hoc). Where both tiers ran, the deterministic one flagged no clean item, and every clean-item flag and every wrong finding came from the model. This is partly by construction: a clean results item was admitted only if the profile and re-execution passed it, and the clean data pools had exact duplicates and impossible rows removed before sampling. On data the union was better than either part; on

results the model's flags already contained the profile's, whose findings its prompt carried. Neither study shows that the model tier scales. The artefacts were small, the model's fabrication findings are hand calculations (two of the 23 rest on wrong ones), and in these runs it flagged no duplicated rows and no shuffled target (0 of 20 each). A model reading is a useful second tier where the artefact fits in the reader's working memory. It is not a substitute for executing the code or computing over the data.

Clean items must be clean by the auditor's standard. In the results and data studies the auditor flagged four defects in our own construction: a card and file that named a column differently, a report claiming a run no file performed, a provenance sentence false for inputs passed as literals, and a unit label contradicting the code's documentation. Each was a correct flag on an item we had built as clean. Two surfaced before the reported readings and were repaired by amendment; the other two surfaced by reading every flag on a clean item, which the registrations required. A fifth construction defect, data files placed outside the audited scope, was not a model finding: it showed up because every verdict came back as an escalation. In the code study the task we wrote named the whole problem when one step was delivered, and a sample of the auditor's flags on correct code objected to exactly that; there we read a sample, not every flag. Without such reading, a false-positive rate can measure the evaluator.

D.6. Limits

1. **Sampling differs between families.** One route is sent temperature 0 and the other no sampling parameter, which bounds how far one union can grow before any difference in ability is measured. Every cross-family union-of- K contrast on general-purpose code inherits this, and it is the confound we would fix first. In the science studies the same-vendor route runs at its default sampling through a command-line route, a different asymmetry.
2. **No matched operating point.** No auditor is compared with another at a matched false-positive rate. Our sweep re-grades archived findings at four deterministic decision rules, under which one family has no point between 3.3% and 34.0% false positives. We built no randomised policy between them, so the paper cannot say whether the families differ at the same operating point.
3. **Interval coverage.** The primary general-code intervals are problem-cluster percentile bootstraps, with cluster counts that depend on the study and stratum (56 problems for the defect population, 143 for the correct sample); the science studies' have their own, and the data study resamples items within dataset. We simulated an analogous single-rate interval, with 600 resamples rather than 10,000 and at the tested rates only. It covers near nominally at the headlines' rates, and below nominal at the two tested rates under 0.10: 0.80 at 0.03 and 0.87 at 0.05 on the defect stratum (§A.3). For the paired contrasts and conditional shares coverage is unmeasured, and no sample size was justified in advance by a detectable-effect calculation.
4. **Substrate.** The substrate is short, self-contained Python functions with executable hidden tests, so nothing here measures auditing a change to a large program with no oracle. Both benchmarks are old enough that memorisation cannot be ruled out.
5. **One generator.** The defect population is one model's failure mode.
6. **Raters.** The residual rubric's raters were two models, one of them the agent that wrote the rubric, which was not blind to the earlier labels. One external label set overlaps eleven of the rated tasks, and the raters' labels agree with it on nine.
7. **The constructed population.** It did not enforce specification-determinedness, and its edits are separable from natural code at 96.7% of the items the probe answered (79.9–97.3% once unanswered items are counted at either extreme) by something the probe does not identify.

Each limit is recorded in the study that carries it, together with the number it would change.

D.7. What we would do next

The sweep re-grades archived findings under four rules and shows that the auditor contrast depends on the rule (§B.2). Re-grading only moves a cut through findings already produced, so it cannot bring every family to a matched operating point: one family has no point under the four deterministic rules between 3.3% and 34.0% false positives. The experiment this paper most needs is to re-ask, instructing each family at graded severity and reading afresh, which can place the families at a common false-positive rate and decide whether the auditor contrast survives there. Matching the sampling parameters across

families would remove the confound above, and reading the cross-vendor family past eight times would turn a statement about our budget into one about a curve.

We ran a first version of the specification experiment, holding the code and the visible tests fixed and varying only the specification (§B.5). Two gaps remain, and they decide what it means. The clarifier inferred the missing rule from failing inputs and was wrong on most instances, so the rule’s correctness was measured, not manipulated. Writing the rule from the oracle for some instances, and a deliberately wrong rule of matched length for others, would separate a specification that settles the question from one that settles it correctly. Its manipulation check also did not show the placebo inert, and a placebo rated no more determining than the original would. Both are cheap, because the code, the tests and the adjudication rubric already exist.

Beyond those: adjudicating more than one reading of the shipped auditor would let a naming rate be compared with a union flag rate at the same K . A second generator would show whether the residual’s character belongs to the benchmark or to one model’s failures. A population of natural defects independently known to be specification-determined would separate the injected population’s artificiality from the property it was built to isolate. And a substrate whose ground truth is not a hidden test suite would test the assumption every ceiling here rests on, that a test suite asked the right question.

E. Limitations in full

Substrate. The general-code studies measure function-level Python tasks from two code benchmarks. The science studies measure one scientific-code benchmark, small results artefacts built from it, and four tabular datasets with synthetic faults. Nothing here establishes that the same ceiling, or the same explanations, hold for another language, for repository-scale changes, for code whose correctness is not decided by an executable suite, or for scientific artefacts larger than a model can read whole. The results and data studies’ faults are injected; the code study’s failures are the generator’s own, judged by a benchmark that contains defective specifications and tests (Hu et al., 2026). One generator wrote every program. The code and data studies read two auditor routes, the results study one, each at its default sampling.

The ground truth is a test suite, and a test suite is not a specification. Hidden suites encode a reference implementation’s behaviour, including behaviour its prose never fixed. In the intervention study reported in §B.5, a generator shown an instance’s failing inputs inferred a behavioural rule that *contradicted* the hidden oracle on 23 of 32 instances. The study cannot say whether that reflects the specifications or the generator’s inference from inputs alone. Either way, it bounds what any recall figure measured this way can mean.

Interval coverage, measured only in part. The primary intervals are percentile bootstraps over problem clusters (over items within dataset in the data study). Coverage was simulated for a *single rate* only. The simulation used the design’s own cluster profiles and an assumed within-problem correlation, with 600 resamples rather than the 10,000 the analyses use and 400 replicates (Monte Carlo standard error 0.01–0.02). Under it, a nominal 95% interval covers between 0.93 and 0.97 at the tested true rates from 0.10 to 0.60, and 0.80 at a rate of 0.03 and 0.87 at 0.05 on the 56-cluster stratum (§A.3). This suggests that small-rate intervals may be narrower than they claim. Review found that two earlier interval constructions in this programme covered 0.416 and 0.075 where 0.95 was claimed.

The validation instrument shares the failure mode under study. Reports were checked by a language model, which is what this paper measures a ceiling on. Studies cleared review at between three and twenty-one rounds, with the round before the last always returning substantive findings. Measurements moved on review in three of the eight general-code studies and in the post hoc counts of two science studies; elsewhere the refusals concerned interpretation, where the instrument is weakest. A further pass under an *attempted* conclusion-suppressed prompt produced six substantive findings that six briefed passes had missed. The supplied artefacts exposed the conclusions anyway, so blinding was not achieved and the comparison is uncontrolled. The defect class this instrument leaves open is the one that looks right to a language model, and we do not know what remains in it.

The reviewer is one of the measured families. gpt-6-astra reviewed every report in this programme and is also an auditor route whose ranking against the others the paper reports. For claims about astra’s relative performance the validation is a self-assessment, and those comparisons should be treated as bracketed until a model that audited nothing here has checked them.

No human rater. Every rating and adjudication in this work was made by a model, one of them the agent that designed the studies and wrote the rubrics (§A.4). A third model rating does not discharge that, and the design that asks for an outside

human rater has not been run.

Budget. Reading depth K , the number of arms, and the size of every intervention population were set by cost rather than by a power calculation, and no sample size was justified in advance by a minimum-detectable-effect argument.

F. Related work in full

Critics that read code. Trained critic models write natural-language critiques of model-written code and catch more bugs than unaided humans (McAleese et al., 2024). That work evaluates on both deliberately inserted and naturally occurring bugs. On code containing natural model errors its critiques are preferred to human critiques in 63% of cases, and it surfaces errors in training data previously rated flawless. It also reports that critics hallucinate bugs, and that a human–model team hallucinates less than the model alone. The natural-defect setting and the precision cost of a louder critic are therefore already its territory, and we claim neither. Its reported quantity is a preference or a per-critique comparison. What we add on this axis is narrow: the union of repeated independent readings as an instrument for locating a limit, ground truth supplied by an interpreter, and the false-positive rate carried beside every recall figure. Aggregating several review passes is known to help. On 1,000 verified pull requests, multi-review aggregation raises F1 by up to 43.67% (Zeng et al., 2025). The motivation is shared with our union-of- K , but there an additional model call synthesises the sampled reviews, while our union counts an instance whenever any reading flags it, so the two estimate different things.

Repeated sampling and coverage. Coverage, the fraction of problems solved by at least one of K samples, rises steeply and then flattens. This is an established regularity of repeated sampling, reported over four orders of magnitude of K and fitted by a power law (Brown et al., 2024). Our union-of- K is that quantity with a flag rule in place of a correctness check, so §B.1 re-measures a known shape, and at $K \leq 8$ our budget is small beside theirs. Our contribution is the reading: the curve is taken as a measurement of a ceiling, reported beside the false-positive rate it costs, and tested against a registered flattening bar that it fails, which is why we report diminishing returns and no limit.

Self-assessment. Two lines of work concern a model judging its own output. Self-correction without external feedback does not reliably improve reasoning and can degrade it (Huang et al., 2024). Models used as judges prefer their own generations, mostly legitimately, since stronger generators are better judges, but harmfully on the instances they themselves get wrong (Chen et al., 2025). Our same-vendor arms are the code-audit case of that literature. Because the routes differ in model, operating point and sampling, we report their contrasts as contrasts between routes, with the false-positive rate beside the recall.

What the hidden suite decides. EvalPlus rebuilt HumanEval and MBPP with larger suites and reported defects in the original reference solutions (Liu et al., 2023); its code carries bespoke oracles for tasks where differential testing against the reference did not apply. Richter & Papadakis (2026) show that models collapse onto a single incorrect reading of an underspecified task, and name twelve MBPP tasks they identified by hand. Eleven fall inside our defect population, and our raters, who labelled them before that paper was fetched, agree on nine. On the test-generation side, the dominant error in generated tests is a wrong expected value rather than an invalid input (Xiong et al., 2023), and generated oracles tend to capture what a program does rather than what was intended (Konstantinou et al., 2024). Our claim is the auditing counterpart: where the prose does not fix the value, a recall number computed against that suite measures the benchmark as much as the auditor.

Consensus as a substitute for ground truth. Several systems validate generated tests without a reference by agreement, through self-consistency, dual execution, or multi-agent discussion for JUnit suites (Taherkhani et al., 2026; Xu et al., 2025). Both measure what survives the filter against a ground-truth implementation: 84–91% of retained tests valid in one (Taherkhani et al., 2026), 0.89–0.93 oracle correctness in the other (Xu et al., 2025). Our measurement differs in unit and in bar. It is the share of wrong tests among retained *failing test–candidate applications*, judged against a preregistered 2% threshold. It is taken over a frozen set of generated tests and candidates, and that set includes candidates already failing their visible tests. Corroboration across independent generations does not meet that validation criterion (§B.5), and wrong tests recur by problem. That recurrence is compatible with the self-consistent errors, the same wrong answer produced again, that degrade the detection methods Tan et al. (2025) evaluate. Our records hold indices, not assertions, so they do not show that the recurring errors are the same one.

Auditing scientific work. On manuscripts, no model exceeds 21.1% recall or 6.1% precision on errata-grade errors and repeated runs rarely rediscover the same errors (Son et al., 2025); the best models detect 46.7% of real paper–code discrepancies (Baumgärtner & Gurevych, 2026). An LLM auditor of AI-scientist projects detects hand-labelled pitfalls

far better with logs and code than with the paper alone (Luo et al., 2025). Pattern-based linting targets methodology bugs in scientific Python (Samsonau, 2026), cleaning agents find row-level errors in corrupted tables and miss errors spanning rows (Bendinelli et al., 2025), and simulation code can run and still solve the wrong equations (Song et al., 2026). SciCode supplies scientist-written tests (Tian et al., 2024), and an expert audit of it found many specifications and tests that reject correct solutions (Hu et al., 2026). Reproduction benchmarks ask agents to recover reported results (Siegel et al., 2024). On constructed statistical workflows, LLM reviewers detect planted flaws almost always but flag 17–50% of correct scripts (Chelate & Heuer, 2026), and Luo et al. already report both true- and false-positive rates. Scientific auditing and paired operating points are therefore not new. What we add is this measurement programme applied to scientific code, and the comparison of the LLM tier with deterministic verification on reported results and data, which we did not find in prior work.

Who audits whom. That an evaluator should not be the model under evaluation is supported by measurements of self-preference in LLM judges against verifiable ground truth (Chen et al., 2025). We do not measure self-preference. A frozen generator’s code audited by routes that differ in model, operating point and sampling cannot separate preference from performance, so we report the same-vendor contrast as a contrast between routes, on one substrate.

Preregistration. Preregistering an evaluation protocol is beginning to appear in machine learning, both as a protocol for experiments with agents (Vaccaro, 2026) and as a defence against tuning prompts, decoding parameters or output format until a desired result appears (Thomas et al., 2026). We count our use of it as method, not contribution.

G. Reproducibility and claims

The repository at <https://github.com/dongzhaohe321418-lab/audit-ceiling-paper> carries the claims ledger (manuscript/CLAIMS.md), with the claims the evidence does not support; the results report of every study the paper draws on; every registered study’s preregistration and numbered amendments (the science studies’ under plan/studies/ai4s/, where the data and results registrations carry errata); every independent review of those reports, verbatim and including refusals; and the derived records (counts, rater labels, hashes and intervals) that the figures and Table 1 are drawn from. A single command, `reproduce.sh`, regenerates Figures 1–4 and Table 1 from those records and checks that each is byte-identical to the copy the manuscript compiles, with no model calls, no network access and no path outside the checkout; it was run from a fresh clone with the home directory unavailable to establish that. The repository’s README maps each claim to its report, its reviews and its records. The corpora are HumanEval (MIT), HumanEval+ and MBPP+ (Apache-2.0), SciCode (Apache-2.0) and four UCI datasets (CC BY 4.0), which permit redistribution. For the three science studies the repository also carries the raw run archive (records/ai4s/runs/): the audited items, every reading’s verdict and the text of its blocking findings, the rating sheets and the logs, so the post hoc adjudications can be checked finding by finding; the final analysis scripts are in records/ai4s/analysis/, and need the harness to run. For the general-code studies it carries each study’s derived records; their raw readings regenerate the records deterministically through the analysis code in the auditor harness, a separate public repository named in the provenance file. Without raw readings, records can be regenerated only by re-running the studies against model endpoints whose outputs are not deterministic.